Segmented Demand and Pricing Strategies for Coffeehouses in Selangor

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Abstract

This study conducts an operational market research analysis with the objective of proposing optimal pricing strategies for a coffeehouse chain in Selangor. Specifically, our focus is twofold: quantifying consumer response to price changes through price elasticity estimation, and deriving optimal prices for different consumer segments segregated according to the type of value obtained from their purchase — utilitarian (Work) and hedonic (Leisure). To perform the analysis, two complementary models were built: a Log-Log Linear Demand model estimated via Two-Stage Least Squares (2SLS) with Hausman instrumental variables for causal inference, and a CatBoost gradient boosting model for flexible demand prediction and price optimisation. These models were selected for their distinct strengths, enabling us to obtain a diverse set of business insights that neither approach could produce alone. Results from our analysis are consistent with economic theory: all ten top-selling beverages exhibit negative and inelastic own-price elasticities, while cross-price relationships are predominantly complementary. Importantly, the two consumer segments display structurally different demand behaviours — Work consumers exhibit smoother, routine-driven demand patterns, whereas Leisure consumers display more volatile and heterogeneous price responses — confirming the validity of the utilitarian–hedonic segmentation framework. Revenue optimisation identifies segment-differentiated optimal prices for three products, while the remaining seven products exhibit approximately segment-invariant optima. These findings provide actionable, segment-specific pricing recommendations for the coffeehouse chain and demonstrate the value of jointly deploying causal econometric and machine learning models in retail pricing analytics.

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1 Introduction

The pricing strategy of any business can significantly influence consumer purchasing behaviour [1] making it a critical micromarketing decision for retailers. Particularly in competitive business environments, such as the beverage coffeehouse industry in Selangor [2], businesses need to establish effective pricing strategies to take advantage of market opportunities and consumer psychology to achieve key business objectives such as market share and profits [3]. The relevance of this Moreover, changes in consummatory behaviour due to fluctuations in prices of various products differ with customer segments [4], making it necessary for businesses to formulate consumer segment specific pricing policies. This motivates us to quantify consumer reactions to price changes by estimating price elasticity and provide deeper insight on optimal prices for each consumer segment. In doing so, we endeavor to assists retailers to tailor pricing strategies that appeal to specific customer segments.

Utilizing transaction-level data from a coffeehouse chain in Selangor consumer, we estimate price elasticities and cross elasticities for the beverages served across 2 consumer segments at the store level. These estimates were acquired by implementing MLR with a Log-Log linear demand model structure. The estimated own product and cross product elasticities captures the price sensitivity and substitution effect of each beverage. This insight builds the foundation for the development of not only pricing strategies, but even other micromarketing strategies like business expansion [5]. To provide further intelligence regarding pricing policies, CatBoost demand models were built. The trained demand models can be exploited to perform a price optimization simulation [6]. Given the output of the CatBoost model, an optimal price that maximizes net revenue can be proposed for each customer segment using grid search. These findings aid coffeehouse retailers to systematically formulate customized marketing strategies for each customer segment propelling them towards their business objectives.

This paper undertakes the following procedure. First, we construct a conceptual framework that guides the identification and investigation of consumer segments in Selangor's coffeehouse market based on existing literature that focuses on price elasticity of demand and discuss the source of data. Second, the demand models developed for each of the top 10 beverage are used to estimate the price elasticities of each consumer segment. In building the demand models, other variables such as method of payment and holidays are also accounted for. The demand models are then thoroughly analyzed and validated, succeeded by a price optimization simulation for the CatBoost model. Finally, we provide insights on different coffeehouse consumer segments in Selangor based on our findings and discuss the limitations of our modelling approach.

2 Background

2.1 Store-Level Research

In the retail and Food and Beverage (F&B) industry, price elasticity of demand is frequently analyzed at the store level since consumer purchasing behaviour can alter according to store-specific characteristics such as competitive environment [7]. Studies also show that another important factor that affects consumer behaviour orientation is store image (often defined by store attributes such as service, atmosphere, layout and even store front displays) [8]. Therefore, we can expect demand levels to differ between the coffeehouse stores in

Selangor. The possibility of observing store specific demand levels indicates that store-level heterogeneity should not be ignored, reinforcing our store-level approach.

MLR is a popular statistical tool used in econometrics to model demand systems [9], even for store-level research. For MLR, store-level heterogeneity is commonly - and often necessarily - modeled using store specific fixed effects. However, this method treats the heterogeneity as nuisance [10], potentially obscuring meaningful patterns in price sensitivity across consumer segments from systematic differences in customer composition and idiosyncratic variation. Early store-level approaches such as Hoch (1988) [7] account for heterogeneity through fixed effects, they treat such variation as unstructured and do not explicitly model systematic differences and idiosyncratic error in consumer behaviour across segments. Random effects or Bayesian models such as that applied by Swarmy (1970) [11] and Allenby & Rossi (1999) [12] respectively allow for explicit modelling of heterogeneity across stores. Nonetheless, they rely on the assumption of independence between price and time invariant store-level un-observables, which is rarely observed in practice [13]. In summary, although store-level demand elasticity analysis is a useful and frequently utilized tool, it can mask unique behaviours of consumer segments.

2.2 Market Segmentation

Consumer price sensitivity may vary across both individual stores and consumer segments. However, retail chains often implement pricing policies at the segment level rather than the store level because highly granular pricing is costly to manage and update [14]. Market segmentation is also widely employed by global coffeehouse chains to manage and exploit differences in consumer price sensitivity. For example, Starbucks has been able to maintain premium pricing despite increasing competition by positioning its stores as "third places" that provide comfortable and quiet spaces for individuals in modern urban environments who may seek locations to work, study, or socialize. These factors motivate our strategy of estimating segment-specific fixed-effects demand models, which can capture store-level heterogeneity while preserving segment-level behavioural differences that may be obscured when a single fixed-effects model is used, as discussed in Section 2.1.

In the context of our dataset, consumers are segmented according to the type of value they are most likely to derive from purchasing coffee: utilitarian and hedonic. The conceptual distinction between these two forms of shopping value was formalized by Barry J. Babin et. al, (1994) [15]. Utilitarian value reflects goal-oriented and functional consumption, whereas hedonic value arises from experiential or enjoyment-based motivations.Based on this framework, stores located in workplace and university environments are classified as utilitarian consumption contexts, where purchases are likily to be driven by productivity needed (eg: studying, or report writing). Conversely, stores in shopping malls and commercial areas are classified as hedonic contexts, where purchases are more likely shaped by leisure and social motivations. This segmentation is adopted because the type of value derived from consumption can influence consumer purchasing behaviour and price sensitivity.

This approach is further supported by evidence from Zaman et al. (2025), who found that consumers in shopping malls and market environments exhibit distinct purchasing behaviours, allowing different retail environments to sustain customer engagement and loyalty.

2.3 Price Optimization

Using demand models for price optimisation—particularly with store-level demand forecasts—is a long-established approach in the literature. Early work by Montgomery (1997) [16] applied price optimisation to store-level supermarket scanner data and showed that profits could be improved through such micro-marketing strategies. This provides the foundation for our own objective of demonstrating how demand-based price optimisation can generate actionable insights from the coffeehouse chain dataset.

More recently, Lee et al. (2023) [17] conducted price optimisation across cereal brands using demand predictions from log-log linear demand systems as well as machine learning models such as LightGBM. Their results showed that machine learning methods consistently outperformed traditional approaches such as ARIMA and multiple linear regression in predictive accuracy. Because these demand predictions serve as direct inputs into the optimisation stage, forecast accuracy is crucial to the credibility of any recommended optimal price. This motivates our use of CatBoost, a gradient boosting framework, to model store-level demand despite its lower interpretability relative to multiple linear regression. Given the high cardinality of store identifiers such as Store_Code, CatBoost's Ordered Target Statistics Section 10 offer a robust non-parametric alternative to one-hot encoding [18], avoiding dimensionality issues and improving the plausibility of the resulting pricing insights.

Particularly, the predict-then-optimise framework adopted in this study is well established in operations research and retail analytics [19]. Its key advantage is flexibility: once a demand model has been estimated, the revenue function can be evaluated directly over a grid of candidate prices without imposing restrictive structural assumptions on its shape. This is especially useful in the present setting, where the PDP and ICE analysis suggests that demand responses may be non-monotonic and segment-specific.

3 Data Set

In this section, we provide a description of the dataset used in our analysis, followed by an outline of the preprocessing measures taken, a brief discussion about the benefits and disadvantages of aggregation to the store level, and an exploratory analysis.

3.1 Data Description

This study uses two years of transaction data from a Selangor coffeehouse chain, covering beverage sales in 2023 and 2024. The dataset includes product information (such as product name, subcategory, temperature, bean type, and milk type where applicable), store information (store code, city, venue, and store type), and transaction information such as payment methods and discounts. A separate dataset on Malaysian public holidays in 2023 and 2024 was also incorporated to account for holiday effects on sales. A more detailed description of the data is provided in Table 17 in Appendix Section 8.2.

The data contains 1,958,127 transactions across 38 products, 18 subcategories, 40 stores, and 20 Selangor cities over 104 weeks. However, not all stores operated throughout the full sample period. Some newly opened stores contained as little as 17 days of records, so 6 stores with fewer than 6 months of data were excluded. Newly opened stores are often highly volatile due to opening effects and operational immaturity [20]; [21], and they also tend to exhibit limited price variation, making price elasticity estimation less meaningful. Since store

fixed effects are used to capture time-invariant heterogeneity, time-invariant store attributes such as city and store type were excluded, except for venue, which is used in segment-level modelling.

Furthermore, product subcategories were also dropped in favour of individual products. This is because sales are highly concentrated within certain subcategories: for example, Cheesecream Latte alone accounts for more than 24.3% of sales within the "Special Latte" subcategory, while the remaining products contribute far less. This suggests that demand is product-specific rather than subcategory-specific. More broadly, the top 10 products account for 73.78% of total sales and 70.82% of total revenue. Consistent with Pareto's Principle [22], focusing on these products captures the main drivers of firm performance and consumer demand.

Moreover, we observe that over time, the fluctuations in unit price differ from store to store, suggesting the presence of store specific pricing and promotional policies. This further confirms our decision to perform a store-level analysis. We verified that for every product in each of the 34 stores, there is sufficient change in price (the minimum price variation observed from all products in the 34 stores was 33%) during the time it was opened until the end of 2024. This is important because limited price movement often leads to imprecise or inconclusive elasticity estimates [23].

3.2 Data Preprocessing

3.2.1 Data Cleaning

First, the two tables (transaction data and holiday information) were combined into one table. Subsequently, we scan for missing values. Missing values were found in "Bean", "Milk", and "Payment Method" columns. The missing values in "Bean" and "Milk" occurred simply because certain beverages do not require coffee beans or milk. For example, Oolong Tea does not require coffee beans, and Espresso does not require milk. The missing values in payment method were combined into the "Others" category, especially since the number of missing values were small.

Next, a number of anomalies were also detected in the dataset. Firstly, we find that for some stores, there are transactions recorded one day before the opening date. These transactions can be explained as soft openings, which are not uncommon for coffeehouses, hence, we adjust the opening date one day earlier to maintain consistency. Additionally, a total of 945 transactions were recorded with negative and zero prices. These records make up less than 0.06% of the overall dataset, and hence, were removed. Similar product names "Dark Chocolate Latte" and "Dark Choco Latte" were also found and combined, especially since we find that there were no more transactions for "Dark Chocolate Latte" after 11th September 2023, and a spike of transactions for "Dark Choco Latte" were recorded after this time point, indicating that they are likely referring to the same product. More details can be found in Table 16 below.

3.2.2 Data Aggregation

To perform a store-level analysis, we aggregate transaction data to the store level. This approach reduces system-wide noise by averaging out stochastic consumer behaviors driven by situational factors (e.g., celebratory purchases) [7]. While store-level aggregation can poten-

tially obscure individual-level signals and introduce aggregation bias [24], regional research indicates that price remains the most prominent driver of food and beverage purchasing behavior in Selangor [25]. Since our dataset exhibits significant price variation at the store level, this aggregation remains appropriate; it preserves the critical relationship between price and demand while filtering out non-systematic individual noise. This aggregation resulted in a 70% reduction in data volume, moving from 1,842,194 raw transactions to 553,006 aggregated daily units.

3.2.3 Feature Engineering

To prepare the dataset for store-level price elasticity modelling, we transform the raw transaction-level data into a store-level panel format. Firstly, we compute net unit price for each product at the store level by dividing the net revenue (total revenue minus all applied discounts) by total quantity sold. This ensures that our price variable captures the "effective price" actually paid by consumers. Next, the long-format transaction data was pivoted into a wide-form structure, with each row representing a unique Store-Date combination, allowing each store to be modelled with its own fixed effect. Additionally, to mitigate the loss of consumer-specific information inherent in aggregation, we engineered category proportions. Specifically, the shares of milk type, bean type, and payment method were measured relative to total store-product volume, providing proxies for changes in product mix and consumer preferences.

When analyzing panel data, temporal dynamics, such as seasonality and long-term trends are treated with great importance for reasons further discussed in Section 4.1.1.2. The seasonal dummies control for periodic demand spikes, while the time trend accounts for long-term growth or contraction in store-level volume. To capture periodic weekly patterns in demand while maintaining parsimony in degrees of freedom, seasonality is controlled for using Fourier terms. Sine and cosine transformations are then constructed as $\sin\left(2\pi\frac{\text{dow}}{7}\right)$ and $\cos\left(2\pi\frac{\text{dow}}{7}\right)$. These terms allow the weekly demand cycle to align more flexibly with observed peaks in consumer activity [26]. The season related variables such as holidays and Ramadan were modeled as dummy variables and temporary structural breaks respectively. In addition, lagged product sales at 1-day and 7-day intervals were engineered to capture short-run persistence in demand. A fuller description of the feature engineering process is provided in Table 18 in Appendix Section 8.3.

3.3 Exploratory Data Analysis

In the following exploratory analysis, we highlight interesting insights gathered from visualizing our data.

3.3.1 Product Specific Sales Distributions

By analyzing the univariate distribution of the quantity of each product from each store, it can be seen that most products have sales and prices with a long right tail.

This suggests the possible need to use log transformation on the response variable (total sales of a product) [17]. To illustrate, Figure 1 clearly demonstrates this skewness price and sales plots for the top 6 most popular products in Store S0007. Appendix Section 8.5 contains plots for all other products as well for Store S0007.

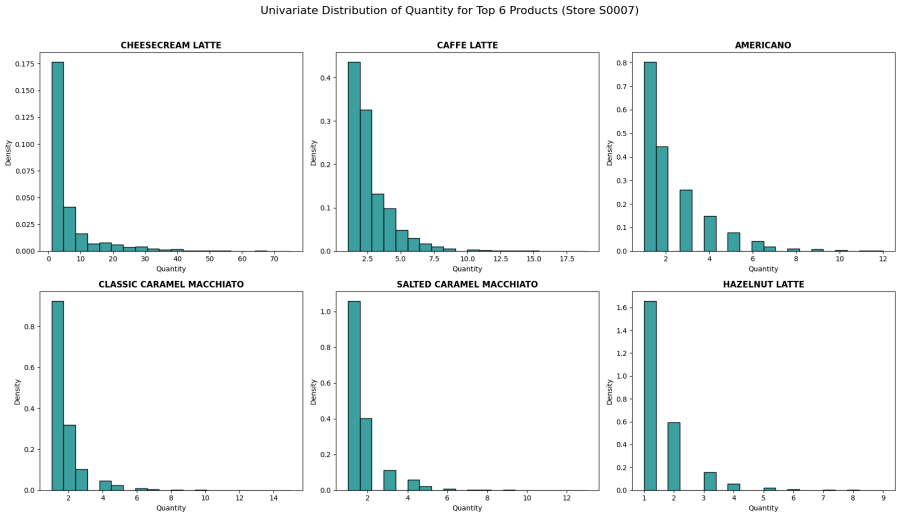


Figure 1: Univariate Distribution of Sales Quantity for Store S0007.

3.3.2 Seasonality & Trend

Demand in retail stores are known to have daily and yearly seasonality [27]; unsurprisingly, this phenomena was observed for the sales of all the stores. Furthermore, by focusing on weekly patterns, it is apparent that the weekly sales dynamics differs for the different venues. More explicitly, looking at figure Figure 2 below - which shows the average sales pattern each day of the week - it can be observed that stores in universities and offices tend to have higher sales during weekdays and drastically lower sales during weekends. Conversely, commercial and shopping mall stores have higher sales during weekends and lower sales during weekdays. These patterns are consistent across almost every store for the entire time series of recorded sales.

This is expected since, for one, individuals usually do not go to offices during weekends but may go to shopping malls. Morover, it is evident that certain office or university stores are not opened during weekends.The observed pattern further supports our method of market segmentation, by demonstrating the possibility of vast differences in purchasing behaviour that occur between consumer segments where vast amounts of individuals purchase out of different value types (hedonic or utilitarian).

A once-a-year decrease in demand is also clearly observed in evident in Figure 3, which shows the aggregated sales of all the products across all coffeehouse stores in Selangor. We note that the plunge in sales corresponds with the Ramadan period, which is highlighted using the shaded blue bars. This pattern is present even when the sales of each product-store pair is analysed individually.

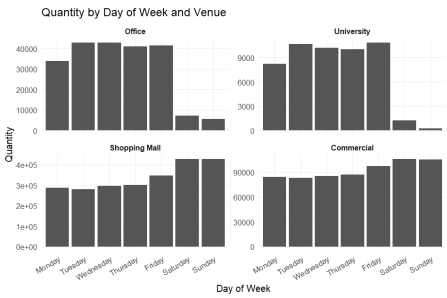


Figure 2: Weekly Sales Volume by Venue.

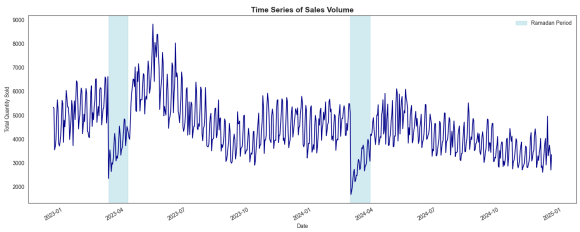


Figure 3: Time Series of Sales Volume.

3.3.3 ACF and PACF

ACF and PACF plots were also plotted as shown in Section 8.6 below. Based on the significant spikes and gradual slope observed in the ACF and PACF plots, we confirm the need to add seasonal dummies, trends and lags in order to account for the trend and overall strong seasonality observed in the sales quantity for each store.

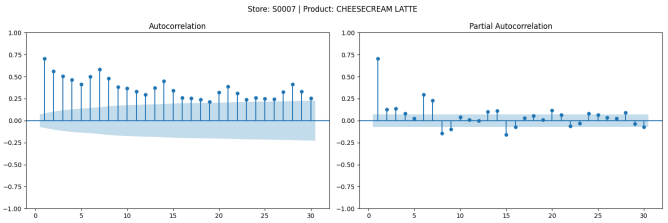


Figure 4: ACF and PACF of Cheesecream Latte in Store S0007.

4 Methodology

This section describes explicitly the process of building Log-Log linear demand models as well as the Catboost model. For each segment, a MLR and Catboost models for each of the top 10 products, giving a total of $2 * 10 = 20$ models for each method. For each method, we detail the process of model specification, variable selection, model building, model diagnostics and model selection. In this study, instead of comparing the capability of each model, we exploit their strengths: for the log-log linear demand models, credible causal inference is prioritized; for Catboost, accurate prediction is prioritized in order to produce accurate price optimization insights.

4.1 Log-Log Linear Demand Model

4.1.1 Model Specification

4.1.1.1 Functional Form Specification

In order to estimate price elasticity, that is, the effect of a 1% increase in price on the percentage increase of quantity, we employ the Log-Log linear demand model. For a market with a homogeneous product, the this model takes the form

$$Q = D(P) = e^{a+\xi_t} P^{-b}. \tag{1}$$

(the plot can be seen in Section 9 Figure 27), where Q represents quantity sold, P the market price, a and b are constants, and ξ_t represents components of demand that are not explained by the model. By taking the natural logarithm of the equation, we get the linear demand equation:

$$\ln Q = a - b \ln P + \xi_t. \tag{2}$$

The own price elasticity (denoted η), is then linear since

$$\eta = \frac{\partial \ln Q}{\partial \ln P} = -b. \tag{3}$$

At this juncture, it is necessary to note that by using this model, a few critical assumptions are made about consumer demand. Firstly, it is assumed that price elasticities of consumers are linear and do not depend on price. Second, we assume that, if $b > 0$, the quantity demanded tends to 0 (but never reaches it) as price tends to infinity, since, taking the inverse demand. Conversely, for $b > 0$, as price tends to zero, demand tends to infinity [9] (refer

to Figure 27). Despite these strong iso-elasticity assumptions, the Log-log linear model is frequently utilized in practice due to its simplicity and interpretability [9]. Since a relatively large amount of data is available to us, it is possible to identify whether or not our data is generated by a Log-Log demand model by performing a formal "Box-Cox" test [9]. Results indicate that a log-log transformation provides the preferred functional form for our products (further details are provided in Appendix Section 8.8), agreeing with our visual analysis in Section 3.3.1.

If a market is such that it can be considered to composed of one singular product, then demand can be modeled using Equation 1. However, based on our dataset, the each coffeehouse offers a variety of 38 products, hence we adopt log linear demand models of the form:

$$\begin{aligned} \ln Q_{1t} &= a_1 + b_{11} \ln P_{1t} + b_{12} \ln P_{2t} + ... + b_{1J} \ln P_{1J} + \xi_{1t} \\ \ln Q_{2t} &= a_2 + b_{21} \ln P_{1t} + b_{22} \ln P_{2t} + ... + b_{2J} \ln P_{Jt} + \xi_{2t} \\ &\vdots \\ \ln Q_{Jt} &= a_J + b_{J1} \ln P_{1t} + b_{J2} \ln P_{2t} + ... + b_{JJ} \ln P_{Jt} + \xi_{Jt} \end{aligned} \tag{4}$$

where $J = 1, 2, ..., 38$ and t represents time in days. Then, we can compute

$$\eta_{ij} = \frac{\partial \ln Q_i}{\partial \ln P_j} = -b_{ij} \tag{5}$$

where η_{ij} is own price elasticity for $i = j$ and cross price elasticity for $i \neq j$. Cross price elasticity measures the effect a change in price of a rival product has on the change in price of the target product.

Note: As discussed in we only focus on the top 10 products in this study

4.1.1.2 Identification and Endogeneity

In Figure 26 in Section 3.3.2, strong weekly and yearly seasonal patterns are observed. These seasonal fluctuations likely reflect external factors such as consumer behavior during periods like Ramadan rather than changes in price. Therefore, the presence of seasonal patterns alone does not imply that they are driven by price fluctuations. Similarly, the slight decreasing trend observed in Figure 3 should not be interpreted as evidence that demand is changing due to price. Inferring causal relationships directly from visual time trends is generally discouraged [9]. This is because these omitted variables may lead to model misspecification, reducing explanatory power and identification of causal effects. Additionally, if the omitted variable is correlated with both the dependent and independent variable, an omitted variable bias could potentially be introduced, implying the existing of an endogenous variable. Thus, we included dummy variables as discussed in Section 3.2.3 to control for seasonality and trend.

While it is possible to identify certain exogenous effects and control for them by including them as variables in our demand model, unobserved variables such as government policy, supply shocks, market competition and income levels are likely to be correlated with both price and demand, causing endogeneity. Another popular cause of price endogeneity is simultaneity, which occurs when random shocks in demand cause suppliers to change product

prices, affecting consumer demand and causing price to be correlated with the error term. For these reasons, price is often considered and endogenous regressor.

4.1.1.3 Fixed Effects & Instrumental Variables

Given the prospect price endogeneity, the OLS assumption of exogeneity (OLS.1 in Section 9.2) required for the Gauss-Markov theorem to hold is likely to be violated, resulting in both biased and inconsistent OLS estimates. Consequently, the usage of fixed effects and instrumental variables (IV) are introduced to address identification and endogeneity issues.

Including control variables and fixed effects, our model for a particular product j takes the form:

$$\ln Q_{st} = a + \sum_{k=1}^{J} b_k \ln P_{kst} + \boldsymbol{X}_{st}\delta + \alpha_s + \xi_{st} \tag{6}$$

Parameter	Description
$\ln Q_{st}$	Log quantity of product j sold in store s at time t.
b_k	Price elasticity of product j with respect to the price of product k. • When $k = j$: own-price elasticity. • When $k \neq j$: cross-price elasticity.
$\ln P_{kst}$	Log price of product k in store s at time t.
$\boldsymbol{X}_{st}$	Vector of time-varying, store-specific control variables.
δ, a	Model coefficients and intercept.
α_s	Store-specific fixed effect, $s = 1, 2, ..., S$.
ξ_{st}	Idiosyncratic demand shock (error term).

Table 1: Parameter List

A good IV for a potentially endogeneous beverage price should be strongly correlated with it but not with the error of the error term of the second stage regression error term. IVs can be utilized by performing 2SLS regression, which is a two stage OLS regression process that begins by regressing price on the IV selected, and subsequently, regressing quantity on the predicted price. Since the IV affects the dependent variable only through the endogenous regressor, using it isolates the exogenous variation in price, removing the bias caused by reverse causality, omitted variables, or other shocks correlated with demand.

In this study, the Hausman-style IV is adopted [28]. Essentially, this IV is obtained by computing the average of the price of the same product in all other stores for a given store. In doing so, it is assumed that demand shocks are store specific and independent across stores and cost shocks are correlated across markets, leaving us with prices changes that are purely due to cost fluctuations (refer to Section 9.4 for details on assumption). These are strong assumptions, and researches recognize that in reality, there is not way to directly test for the existence of endogeneity, much less eliminate it entirely [29]. However, by

adopting Hausman's IV, we create a situation whereby observing a causal effect is possible.

2SLS Hausman IV Model Specification

The structural equation for the demand of product j across stores s at time t is:

$$\ln Q_{st} = a + \sum_{k=1}^{J} b_k \ln P_{kst} + \boldsymbol{X}_{st}\delta + \alpha_s + \xi_{st} \tag{7}$$

For each endogenous price $\ln P_{kst}$, we use the Hausman instrument $Z_{kst} = \overline{\ln P}_{k,-s,t}$, which is the average price of product k in all stores other than s at time t:

$$\ln P_{kst} = \gamma_0 + \gamma_1 \overline{\ln P}_{k,-s,t} + \boldsymbol{X}_{st}\Pi + \alpha_s + \nu_{st} \tag{8}$$

We substitute the stochastic endogenous prices with the predicted values $\widehat{\ln P}_{kst}$ obtained from the first stage:

$$\ln Q_{st} = a + \sum_{k=1}^{J} b_k \widehat{\ln P}_{kst} + \boldsymbol{X}_{st}\delta + \alpha_s + \xi_{st} \tag{9}$$

The assumptions required and estimator for 2SLS regression are specified in Section 9.4 and will be discussed further in Section 4.1.3 and Section 4.1.4.

4.1.2 Variable Specification

Category	Variable	Description
Price	LOG_UP	Log of unit price; endogenous variable capturing own- and cross-price effects
Control	RAMADAN	Indicator for Ramadan period
	PAY_G	Share of GrabFood payments
	PAY_C	Share of card payments
	PAY_A	Share of in-house app payments
	LAG_1	Lagged quantity at $t-1$ (short-run persistence)
	LAG_7	Lagged quantity at $t-7$ (weekly pattern)
	DOW_SIN	Sine transformation of day-of-week (weekly seasonality)
	DOW_COS	Cosine transformation of day-of-week (weekly seasonality)
Instrument	LOG_LOO_UP	Log of leave-one-out price (Hausman instrument based on other stores)
Fixed Effects	STORE_CODE	Controls for time-invariant store-level heterogeneity
Segment	PURPOSE	Consumer segment indicator (Work vs Leisure)

Table 2: Classification of variables used in the 2SLS demand model.

Table 2 above provides a list of features that were considered for our model. The following two sections details the process of selecting these features and evaluating our instrument variables.

4.1.3 Model Development

4.1.3.1 Instrumental Variable Validation

Initially, 2SLS regression was performed with all the control variables and own and cross price elasticities. The partial R^2 values from the First Stage regressions of each IV was then analyzed to understand the proportion of variation in the endogenous variable explained by the instruments, after the variation from common exogenous variables was partialled out [30]. The observed partial R^2 values were high (≈ 0.8), indicating that the instruments are strongly correlated with price, satisfying Assumption 2SLS.3 in Section 9.4. To formally assess instrument strength, a Stock and Yogo (2005) weak instrument test was performed.

The resulting First-stage F-statistic exceeded the critical threshold for 10% relative bias by a sizeable margin, allowing for the rejection of the null hypothesis that the instruments are weak [31].

Product	CHEESE CC	VANILLA V	AMERI A	CAFFE CAF	PECAN P	HAZEL H	SALTED SCM	MATCHA M	COCOA CO	CLASSIC CCM
Partial R²	0.789	0.782	0.656	0.722	0.742	0.734	0.851	0.786	0.741	0.747
Other Partial R²	0.752 (P)	0.736 (CAF)	0.799 (CC)	0.711 (CCM)	–	0.711 (A)	0.695 (CO)	0.804 (V)	0.817 (CC)	0.743 (P)
F-Stat	2612 2117 (P)	2684 2086 (CAF)	1650 3437 (CC)	2262 2139 (CCM)	3881	2007 1795 (A)	4175 1672 (CO)	2488 2789 (V)	2080 3257 (CC)	1503 1472 (P)

Table 3: True First-Stage Partial R² by Endogenous Regressor (Work)

Product	CHEESE CC	VANILLA V	AMERI A	CAFFE CAF	PECAN P	HAZEL H	SALTED SCM	MATCHA M	COCOA CO	CLASSIC CCM
Partial R²	0.807	0.836	0.784	0.867	0.794	0.858	0.815	0.783	0.804	0.808
Other Partial R²	0.772(M)	0.875(CC)	0.859(CC)	0.793(V)	0.810(CO)	0.784(P)	0.848(CCM)	–	0.828(CC)	0.782(CO) 0.827 (CC)
F-Stat	38982 31445 (M)	42208 58284 (CC)	34490 58081 (CC)	61134 35864 (V)	32449 36016 (CO)	51906 31361 (P)	40256 50834 (CCM)	62443	34996 41034 (CC)	25422 21693 (CO) 28904 (CC)

Table 4: True First-Stage Partial R² by Endogenous Regressor(Leisure)

4.1.3.2 Feature Selection

After verifying the strength of our IVs, the model underwent refinement to ensure parsimony. Cross-price elasticities were evaluated based on their economic magnitude; any cross-product coefficient with an absolute value less than 0.1 ($|\eta_{ij}| < 0.1$) was removed from the system. These negligible values indicate a lack of meaningful substitution or complementarity between those specific product pairs, and their removal preserves degrees of freedom without sacrificing the model's explanatory power as recommended by Krämer, Walter, [32]. Finally, a sensitivity analysis of the control variables was conducted by dropping them and observing the effect on the target and rival price coefficients. It was identified that dropping the trend term for the work segment resulted in a change in sign for the own product elasticities. Since we observed a mild decreasing trend in the overall time series, it can be concluded that removing trend results in an omitted variable.

4.1.4 Model Diagnostics

In this section, we verify whether our models satisfy the critical assumptions underlying 2SLS regression. To test for homoscedasticity (Assumption 2SLS.5), residual plots were examined. Based on these illustrations, the potential existence of mild heteroscedasticity is acknowledged (see, for example, Figure 5). Furthermore, ACF and PACF plots were analyzed to detect temporal dependencies. Despite the explicit inclusion of lags in the structural model, certain store-product combinations suggested the possible presence of serial correlation (see Figure 5).



Figure 5: Scatterplot of residuals suggesting mild heteroscedasticity.
Left: Salted Caramel Macchiato(Work). Right: Salted Caramel Macchiato(Leisure)

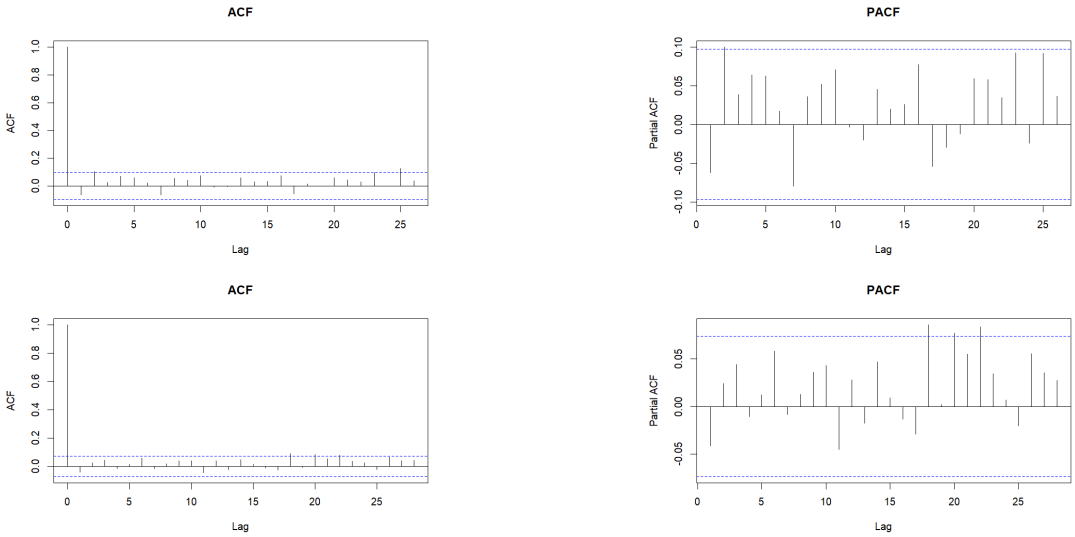


Figure 6: ACF and PACF diagnostics. Top: Salted (Work). Bottom: Hazelnut (Leisure).

It is important to note that the 2SLS estimator remains consistent under Assumptions 2SLS.1–2SLS.4 even in the presence of these violations. However, the classical 2SLS standard errors are no longer valid for hypothesis testing, as the estimator's asymptotic normality requires a correctly specified variance-covariance matrix. To ensure valid inference, Newey-West standard errors were applied to correct for both remaining autocorrelation and heteroskedasticity (HAC) in the residuals. Additionally, we assessed multicollinearity to ensure the regressors satisfy the full-rank condition (Assumption 2SLS.3).

Finally, to assess the robustness of our models, we conducted an outlier sensitivity analysis. Specifically, this analysis was performed for the Cheesecream Latte models, which demonstrated heavy left tails in the quantity distribution for both segments (Figure 7). Observations in the extreme tail was removed using a trimming threshold (the 5th percentiles of the price and quantity distributions), and the models were re-estimated to verify that the primary results were not driven by extreme observations. Following this re-verification, we observe minimal change in the value of the coefficients, demonstrating the robustness of our model to outliers. Consequently, the original specification was maintained for final inference.

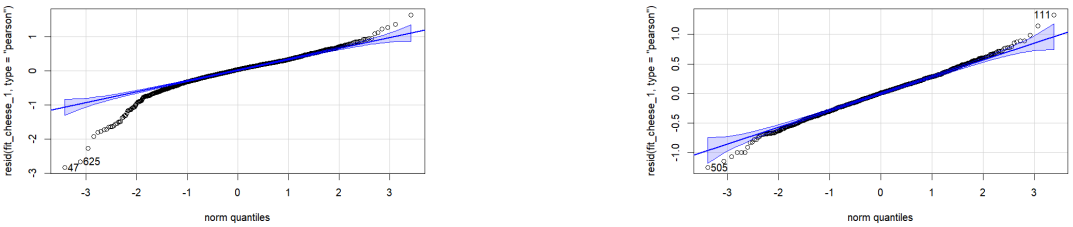


Figure 7: QQ Plot of Cheesecream Latte before (left) and after (right) trimming 5th percentile tail

4.2 CatBoost Demand Model

To complement the log-log linear demand system, we estimate a nonlinear demand mode using CatBoost, a gradient boosting framework that constructs ensembles of symmetric (oblivious) decision trees [33]. CatBoost is well suited to the present store-level daily panel because the dataset combines continuous variables — prices, discounts, lagged quantities, and time controls — with high-cardinality categorical variables such as store identifiers and purpose labels. While the log-log linear model is attractive for its direct interpretability, it imposes a restrictive functional form and requires interactions and nonlinearities to be specified manually. CatBoost instead learns these relationships flexibly from the data. Its ordered boosting procedure further reduces the prediction shift that arises from target leakage in standard gradient boosting implementations [33]; further details on ordered boosting are provided in Section 10.1.

4.2.1 CatBoost Algorithm

Figure 8 illustrates a single **symmetric** tree of depth 3. The defining property of symmetric (oblivious) trees is that the same **splitting condition is applied to every node** at a given depth — for instance, all nodes at depth 2 in Figure 8 split on the same feature (Lag1_Sales $<$ 1.10), regardless of the path taken from the root. This constraint **reduces overfitting** and accelerates inference, as each observation's leaf assignment can be determined by a simple bitwise operation on the split outcomes rather than sequential node-by-node traversal [33].

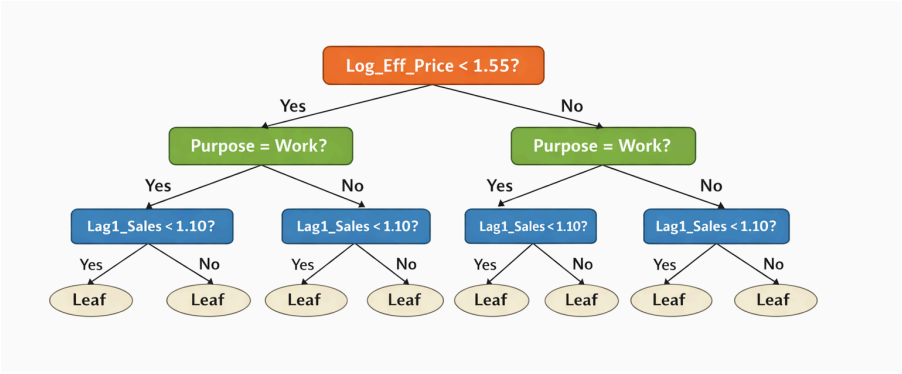


Figure 8: Catboost Symmetric tree with depth = 3

This structure allows CatBoost to approximate nonlinear functions and higher-order interactions without requiring them to be specified explicitly in advance. Read Section 10 for more detailed formulas on catboost.

4.2.2 Model Specification

This CatBoost model is estimated on the log-transformed demand variable, $\ln\left(Q_{\{ist\}}\right)$ where $Q_{\{ist\}}$ denotes quantity sold for product i in store s on day t .The feature vector includes own and cross prices, discount indicators, lagged quantities, store identifiers, purpose labels, and calendar controls — mirroring the covariates used in the log-log specification. Formally, the fitted model may be written as

$$\ln\left(Q_{\{ist\}}\right) = \hat{f}\left(x_{\{ist\}}\right) = \sum_{\{m=1\}}^{M} f_{m\left(x_{\{ist\}}\right)} \tag{10}$$

where $f_{m\left(x_{\{ist\}}\right)}$ denotes a symmetric regression tree and M is the total number of boosting iterations.

4.2.3 Feature Construction

For each target beverage, the CatBoost model uses store-day features that broadly mirror the linear demand model. These include the target product's `log effective price`, `lagged log sales` at horizons of 1, 2, and 7 days, calendar and seasonal controls (cyclical day-of-week encoding, month factors, a linear time trend), `holiday` and `Ramadan indicators`, and payment composition shares. A count of daily active stores within the same purpose segment (`Purpose_Stores_Active`) proxies for same-segment competitive intensity, while the variable `purpose` was created to capture the segment structures. Store-level heterogeneity is also incorporated directly through the store identifier.

All categorical features are passed to CatBoost becomes part of the tree-based learning process through CatBoost's ordered target statistics encoding. Explained in Section 10.4. This allows the model to capture persistent store-level differences in baseline demand and, where supported by the data, more complex interactions between store identity and other predictors.`store_code`.

4.2.4 Rival Product Selection and Cross-Product Effects

To capture substitution patterns within the beverage menu, rival-product prices are included in the CatBoost model. Since including every product in the menu as a rival for each target beverage would be neither practical nor stable, a data-driven rival-selection rule is adopted.

All products are first ranked according to total quantity sold across the full sample. For a given target beverage, the target product itself is excluded from this ranking, and the highest-selling remaining products are treated as rival candidates. These candidates are then screened so that only beverages with a sufficient number of positive-price observations in the target-product estimation sample are retained. For each selected rival, the log of rival price is included as a predictor, allowing the CatBoost model to capture flexible cross-product effects without imposing a strictly linear cross-elasticity structure.

Full features table was included in the appendix Section 10.2

4.2.5 Model Training and Hyperparameter Tuning

The model is estimated using a root mean squared error (RMSE) loss function. Hyper-parameters are selected via Bayesian optimisation using the Optuna framework [34]), which

employs a Tree-structured Parzen Estimator (TPE) sampler to search the hyperparameter space more efficiently than exhaustive grid search.

Because the data have a temporal structure — the feature set includes lagged demand variables and a time trend — standard random k-fold cross-validation is inappropriate for hyperparameter selection, as it would allow future observations to leak into the training set through the lag features [35]. Instead, we adopt a *temporal holdout* validation strategy. For each product–purpose model, the training data are sorted chronologically and split at a date cutoff such that the first approximately 80% of unique dates form the training partition and the remaining 20% form the validation partition. The split is performed on unique dates rather than row counts to ensure that all observations from a given calendar day remain together, preventing same-day information leakage.

Within each Optuna trial, a CatBoost model is fitted on the training partition and evaluated on the validation partition. CatBoost's built-in early stopping mechanism monitors validation RMSE and terminates training if no improvement is observed within 50 boosting rounds, ensuring that unpromising hyperparameter configurations are abandoned quickly. Table 5 summarises the hyperparameter search space explored by Optuna.

Hyperparameter	**Range**	**Scale**	**Description**
Tree depth	4 – 8	Integer	Maximum depth of each boosting tree
Learning rate	0.01 – 0.15	Log	Shrinkage applied to each boosting step
L_2 leaf regularisation	1.0 – 15.0	Log	Penalty on leaf weights to reduce overfitting
Boosting iterations	200 – 800	Integer*	Maximum number of trees before early stopping
Minimum data in leaf	5 – 50	Integer	Minimum observations required per terminal node
Bagging temperature	0.0 – 1.0	Continuous	Controls randomness of Bayesian bootstrap weighting

Table 5: Optuna hyperparameter search space for CatBoost tuning. Each model undergoes 40 trials using TPE (Tree-structured Parzen Estimator) sampling. *Sampled in steps of 100.

A total of 40 Optuna trials are conducted per model, which, given the Bayesian sampling strategy, typically covers the effective search space more thoroughly than a comparable grid search of several hundred deterministic combinations.

Once the optimal hyperparameters are identified, the *final model is refit on the full dataset* — including both the training and validation partitions — using those hyperparameters. The temporal split is used exclusively for hyperparameter selection; the production model is trained on all available data to maximise the information available for learning. **GPU training** is employed throughout to improve computational efficiency across the product-level models.

4.2.6 Model Selection: Per-Purpose versus combined Specification

Two modelling strategies were compared to determine how best to incorporate the Work–Leisure segmentation. The first, referred to as the *per-purpose model*, splits the data by store purpose and fits a separate CatBoost model for each product–purpose combination. The second, referred to as the *combined model*, trains a single CatBoost model per product on the combined `Work + Leisure` dataset, incorporating purpose as a categorical feature and including `own-price $times$ purpose` and `rival-price $times$ purpose` interaction terms so that the model can, in principle, learn segment-specific effects within a unified framework.

To ensure a fair comparison, the combined model's predictions were evaluated on the same segment-specific subsets used by the per-purpose models: combined-model predictions were filtered to Work-only and Leisure-only observations, and metrics (RMSE, MAE, R^2, MAPE) were computed within each subset.

4.2.7 Model Interpretation and Diagnostics

Model interpretation relies on SHAP (SHapley Additive exPlanations), which assigns each feature an importance value for each individual prediction based on Shapley values from cooperative game theory (Lundberg and Lee, 2017). SHAP summary plots reveal both the ranking and directionality of feature effects. In addition, Partial Dependence Plots (PDP) and Individual Conditional Expectation (ICE) plots are used to visualise the average and store-level price–quantity relationships implied by the fitted model, providing a complementary view of price sensitivity that is directly comparable to the demand curves estimated in the linear framework.

Additional PDP, ICE, and SHAP is provided in Appendix Section 10.

Since CatBoost is a nonlinear ensemble model, its performance is not interpreted through a small set of coefficients in the same way as the linear demand system. Instead, model evaluation is based on predictive performance metrics and post-estimation interpretation tools. In particular, model fit is assessed using RMSE, MAE, and R^2 on the log-quantity scale, together with RMSE, MAE, and MAPE on the original quantity scale. These metrics allow predictive accuracy to be compared across products and across **combined** and **purpose-specific specifications**.

4.2.8 Revenue Optimization

To translate the fitted demand models into actionable pricing recommendations, we adopt a *predict-then-optimise* framework. This two-stage approach — first estimating a demand model, then using it as a simulator to evaluate candidate prices — is standard in the revenue management literature and has been applied in retail pricing contexts by Ferreira, Lee, and Simchi-Levi (2016)[19], who used nonparametric demand predictions from regression trees as inputs to a multi-product price optimisation at the online retailer Rue La La.

The procedure is as follows: For each product–purpose combination, a price grid is constructed from the 5th to the 95th percentile of observed effective prices, discretised into 40 evenly spaced candidate prices. This range restriction ensures that the optimisation searches only over prices for which the model has adequate training support, **avoiding extrapolation** into regions where the fitted demand surface is unreliable. This design choice avoids spurious extrapolation but implies that the reported optima are conditional on the range of prices the chain has historically set.

For each candidate price p on the grid, the CatBoost model is used to generate a counterfactual demand prediction: the log price feature is set to $\ln(p)$ for all store-day observations in the segment while all other features (lagged sales, day-of-week, store identity, payment shares, rival prices) are held at their observed values. The model then predicts $\hat{Q}_{ist}(p) = \exp\big(\hat{f}(x_{st}; p)\big)$ for each observation, and the total predicted revenue at price p is computed as

$$R(p) = p \cdot \sum_{s,t} \hat{Q}_{ist}(p). \tag{11}$$

The detailed formula will be included in the appendix Section 10.8

Note: The resulting optimal prices can be validated using ICE curves or SHAP analysis to confirm consistency with the model's inferred demand sensitivities. In both cases, the target product's price is varied over a grid while other covariates are held at their observed values. However, whereas the PDP summarizes the model-implied demand response by plotting predicted quantity against price, the revenue optimisation transforms these predictions into predicted revenue by multiplying predicted quantity by the candidate price. The optimal price is then identified as the price that maximizes this model-implied revenue curve.

5 Results and Discussion

5.1 Log-Log Linear Demand Model

5.1.1 2SLS Regression

Note: Significance levels: . p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001.

Product	CC	V	A	CAF	P	H	SCM	M	CO	CCM
Own Elasticity	−0.219 (0.199)	−0.354** (0.128)	−0.125 (0.302)	−0.615*** (0.133)	−0.388** (0.143)	−0.599*** (0.165)	−0.859*** (0.167)	−0.617*** (0.160)	−0.318. (0.168)	−0.733*** (0.163)
Cross Elasticity	−0.102. (0.055) (P)	−0.498** (0.161) (CAF)	–	−0.157. (0.092) (CCM)	–	–	−0.211** (0.078) (M)	−0.229** (0.088) (V)	–	−0.258** (0.086) (P)

Table 6: Demand Elasticities and Instrument Strength (Work)

Product	Rama	Pay_A	Pay_C	Pay_G	*t*	Lag1	Lag7	sin	cos	Adj R^2	W. R^2	W-H(p)
CHEESE	−0.396***	–	−0.032	–	−0.0009***	0.153***	0.091***	0.116***	−0.051***	0.461	0.380	0.225
VANILLA	−0.504***	–	−0.036	–	−0.001***	0.124***	–	0.188***	−0.104***	0.440	0.188	< 0.001
AMERI	−0.152.	–	–	–	0.0002.	0.210***	0.143***	0.349***	−0.006	0.696	0.279	< 0.001
CAFFE	−0.441***	–	−0.233	–	−0.0006***	0.149***	0.168***	0.266***	−0.031*	0.739	0.318	0.0004
CLASSIC	−0.751***	–	−0.586**	–	−0.0007***	0.116***	0.131***	0.155***	−0.103***	0.390	0.240	< 0.001
PECAN	−0.823***	–	−0.684**	–	−0.0007***	0.098**	0.057*	0.163***	−0.052**	0.449	0.161	< 0.001
HAZEL	−0.460***	–	−0.196	−0.122	−0.0012***	0.155***	0.115***	0.191***	−0.001	0.414	0.285	< 0.001
SALTED	−0.673***	0.397	–	0.223	−0.001***	0.088**	0.050.	0.122***	−0.099***	0.334	0.233	0.869
MATCHA	−0.454***	–	−0.538*	–	−0.0005***	0.160***	0.100***	0.155***	−0.001	0.477	0.164	0.012
COCOA	−0.645***	0.158	–	–	−0.0006***	0.117***	0.092**	0.195***	−0.076***	0.316	0.147	< 0.001

Table 7: Two-Stage Least Squares (2SLS) Estimates of Demand with Fixed Effects (Work Segment)

Product	CC	V	A	CAF	P	H	SCM	M	CO	CCM
Own Elasticity	−0.435*** (0.096)	−0.680*** (0.068)	−0.199** (0.064)	−0.595*** (0.060)	−0.397*** (0.074)	−0.637*** (0.079)	−0.499*** (0.091)	−0.545*** (0.049)	−0.335*** (0.055)	−0.620*** (0.081)
Cross Elasticity	-	−0.631*** (0.139) (CC)	−0.358*** (0.108) (CC)	−0.135*** (0.024) (V)	−0.182*** (0.039) (CO)	−0.242*** (0.046) (P)	−0.339*** (0.058) (CCM)	–	−0.659*** (0.152) (CC)	−0.170*** (0.032) (CO) −0.706*** (0.127) (CC)

Table 8: Demand Elasticities and Instrument Strength (Leisure)

Product	Rama	Pay_G	Pay_C	Pay_A	Lag1	Lag7	sin	cos	Adj R^2	W. R^2	W-H(p)
CHEESE	−0.221***	–0.119*	– 0.114***	−1.06***	0.377***	0.240***	−0.104***	−0.085***	0.669	0.455	8.664e-8
VANILLA	−0.232***	−0.517***	−0.326***	−1.715***	0.140***	0.149***	−0.146***	−0.035***	0.349	0.203	< 2.2e-16
AMERI	−0.094***	–	−0.188***	−0.334***	0.146***	0.164***	−0.109***	−0.022***	0.631	0.097	< 2.2e-16
CAFFE	−0.194***	–	–	−0.757***	0.189***	0.230***	−0.112***	−0.029***	0.663	0.275	5.299e-15
PECAN	−0.291***	−0.681***	−0.470***	−1.200***	0.173***	0.150***	−0.148***	−0.055***	0.377	0.186	< 2.2e-16
HAZEL	−0.191***	−0.447***	–	−1.547***	0.176***	0.172***	−0.148***	−0.036***	0.370	0.234	< 2.2e-16
SALTED	−0.202***	–	−0.518***	−1.531***	0.207***	0.212***	−0.161***	−0.050***	0.426	0.286	< 2.2e-16
MATCHA	−0.219***	−0.193**	−0.445***	−1.243***	0.142***	0.155***	−0.144***	−0.051***	0.366	0.163	< 2.2e-16
COCOA	−0.238***	−0.249***	−0.256***	−1.441***	0.121***	0.123***	−0.165***	−0.033***	0.295	0.134	< 2.2e-16
CLASSIC	−0.266***	–	−0.363***	−1.616***	0.222***	0.191***	−0.140***	−0.050***	0.473	0.285	< 2.2e-16

Table 9: Two-Stage Least Squares (2SLS) Estimates of Demand with Fixed Effects (Leisure Segment)

5.1.2 OLS Regression

Product	CC	V	A	CAF	P	H	SCM	M	CO	CCM
Own Elasticity	−0.045 (0.201)	−0.225. (0.129)	0.372 (0.270)	−0.440*** (0.127)	−0.155 (0.125)	−0.057 (0.270)	−0.838*** (0.159)	−0.108 (0.181)	−0.065 (0.150)	−0.410. (0.245)
Cross Elasticity	−0.045 (0.053) (P)	−0.375** (0.144) (CAF)	–	−0.103 (0.089) (CL)	–	−0.040 (0.270) (A)	−0.211** (0.078) (M)	−0.012 (0.129) (V)	–	−0.084 (0.100) (P)

Table 10: OLS Demand Elasticities (Work)

Product	CC	V	A	CAF	P	H	SCM	M	CO	CCM
Own Elasticity	−0.230* (0.097)	−0.456*** (0.067)	0.008 (0.070)	−0.547*** (0.076)	−0.237*** (0.066)	−0.450*** (0.078)	−0.289*** (0.086)	−0.297*** (0.055)	−0.121* (0.050)	−0.404*** (0.101)
Cross Elasticity	-	−0.681*** (0.132) (CC)	−0.339*** (0.096) (CC)	−0.093*** (0.025) (V)	−0.117** (0.037) (CO)	−0.215*** (0.043) (P)	−0.361*** (0.054) (CCM)	–	−0.639*** (0.133) (CC)	−0.115*** (0.032) (CO) −0.783*** (0.118) (CC)

Table 11: OLS Demand Elasticities (Leisure)

5.1.3 Model Interpretation Preliminaries

Firstly, it should also be noted that the R^2 values obtained are essentially meaningless for 2SLS regression and does not measure the goodness of fit of the estimator [36]. In fact, the R^2 values obtained can even be negative. This is because the R^2 is defined using the original regressors X rather than the instrumented values $\hat{X}$. refer to Section 9.5.3. Because the 2SLS estimator $\hat{\theta}_{\text{2SLS}}$ is calculated to minimize the residuals of the projected values and not the actual observed values, it is mathematically possible for the Residual Sum of Squares to exceed the Total Sum of Squares:

$$RSS > TSS \rightarrow \frac{RSS}{TSS} > 1 \rightarrow R^2 < 0. \tag{12}$$

For this reason, Wooldridge notes that, if the objective was to achieve a good "fit", the OLS model should be used [36]. On the other hand, if inference was the objective and there is an endogenous regressor, the 2SLS estimates can be more consistent and unbiased compared to OLS at the cost of efficiency. By comparing Table 6 vs Table 10 and Table 8 vs Table 11 we can see that the standard errors for the OLS model are generally smaller than that of 2SLS for price elasticity.

Additionally, statistical significance can be misleading when sample sizes are large. Consequently, we re-emphasize the need to consider the economic magnitude [32] and consistency of our estimates and compare them with our data visualizations as well.

5.1.4 OLS vs 2SLS regression

The Wu-Hausman test for endogeneity yielded p-values consistently below 0.01 [37]. This statistically significant divergence between the OLS and 2SLS estimates confirms that price is endogenous within this demand system; if price were exogenous, the OLS and IV estimates would be numerically similar. Consequently, the 2SLS framework is necessary to provide consistent estimates of price elasticity. Although the p-values could be inflated due to large sample sizes, given the extremely small p-values obtained (refer to Table 23) it is plausible for the null hypothesis to be rejected even with a smaller sample size.

Besides, these observations can be compared to the estimates obtained from an OLS regression built on identical features, where we note that the OLS frequently underestimates the price elasticities for most products, which is anticipated [38]. This is especially obvious for Americano, where we note that the 2SLS own-price elasticity estimates from 2SLS regression (Table 8 and Table 6) are -0.125 and -0.199 respectively (price inelastic) while the OLS estimates are 0.372 and 0.008 respectively (positive elasticity and alsmot perfectly inelastic). This discrepancy demonstrates the possibility of obtaining inaccurate causal inferences if endogeneity is ignored.

5.1.5 Price Elasticity of Demand

Before evaluating the price elasticity estimates, we first establish that, according to economic principles, if the absolute value of price elasticity is less than 1, the product is considered inelastic, however, if it is more than one, it is considered elastic. All the interpretations and inferences regarding the coefficient of a variable are also made assuming all other variables are kept constant.

5.1.5.1 Own-Price Elasticity

Generally, beverages are considered normal goods, since demand usually decreases as price increases, and hence, price elasticities are expected to be negative [39]. Based on the Tables above, we note that all the products demonstrate negative price elasticities for both work and leisure segments, indicating that our estimates align with economic theory. Even the non-statistically significant own price elasticities remain negative one standard error away from the estimate. It is also evident that all the products in both segments are price inelastic, demonstrating own-price elasticities of less than one, implying that a 1% increase in price decreases demand by less than 1%. In other words, customers will be less reactive to increases in price. This can be an indication of the coffeehouse chain's market dominance. Possibly due brand loyalty or uniqueness of products.

5.1.5.2 Cross-Price Elasticity

Moreover, all of the statistically and economically significant rival products for each of the top 10 products are complements (cross price elasticity is negative). This is consistent with general consumer purchasing behaviour when it comes to a competitive market such as the coffehouse market in Selangor. Especially since coffeehouses are often used as places for meetings and gatherings, an increase in price of one product can cause groups of people to boycott a store completely, causing a decrease in demand for other products to decrease.

Analysis of the complementary goods for the Leisure and Work segments (Table 8) reveals that most of the top 10 products are linked to the Cheesecream Latte as a complement. This aligns with its status as a bestseller; customers visiting primarily for a Cheesecream Latte

often purchase additional items, meaning a price-driven drop in Cheesecream Latte demand likely pulls down sales for associated products. On the other hand, this pattern is not present for the work segment. This is unsurprising and in line with the method of segmentation - according to hedonic and utilitarian value. Consumers at offices or universities tend to have utilitarian value, purchase out of need, for example, for caffein.

Additionally, all cross-elasticities remain below 1, indicating an inelastic relationships across all products. Within the work segment, the Peacan Crunch Latte exhibits the smallest cross-price elasticity. Specifically, the cross-price elasticity for the Peacan Crunch Latte relative to the Cheesecream Latte is around 0.1. While statistically significant, the economic magnitude is still relatively small: for every 1,000 cups of Cheesecream Latte sold, a 1% increase in price of Matcha Latte, 0.1% decrease in demand represents less than a single cup, demonstrating limited practical impact on demand. A similar arguement can be used to interpret all other price elasticity estimates.

Overall, implies that, while demand is mostly price inelastic, increases the price of their best seller causes a decrease in demand across many of the top 10 products.

5.1.6 Deterministic Seasonality

Finally, we note that the Ramadan coefficients are almost all negative, which agrees with our visualization in Figure 25. There is also a systematic difference between the sin and cos variables used to control for seasonality: the sin coefficients are all positive for the work segment but all negative for the leisure segment. This reflects the difference in weekly demand patterns for these two segments as discussed in Section 3.3.2. While the previously observed mild trend was significant in the work segment, it was not significant in the leisure segment. The "mildness" of the trend was captured by the small coefficients corresponding with t.

5.2 CatBoost Demand Model

5.2.1 Hyperparameter Tuning Outcomes

The **Optuna tuning** procedure converged efficiently across all 30 models (10 products × 3 specifications: combined, per-purpose Work, per-purpose Leisure), completing **40 Bayesian trials per model**. Early stopping was triggered in every case, with the effective number of boosting iterations consistently falling below the upper bound requested by the sampler — confirming that the search space was **appropriately calibrated** .

Several patterns emerge from the tuned hyperparameters. **Work-segment** models generally selected **deeper trees** (depth 5–7), **lower learning rates** (0.03–0.10), and **higher minimum leaf sizes** (22–48) relative to their **Leisure** counterparts,Leisure models, by contrast, tended toward shallower trees (depth 4–5) and could tolerate lower regularisation,reflecting the smaller sample size and the need for stronger regularisation to avoid overfitting.

The **combined models** occupied an intermediate position, but their tuned configurations often resembled those of the Leisure segment — consistent with the fact that Leisure observations dominate the combined loss function and therefore drive the gradient updates.

The total tuning time across all 30 models was approximately 120 minutes on a single GPU, with individual models requiring 2–7 minutes depending on dataset size and the depth of the search.

5.2.2 Model Diagnostics and Fit

Table 13 reports the primary goodness-of-fit metrics for the per-purpose (PP) and combined models, evaluated on matching segment-specific subsets to ensure a fair like-for-like comparison. **RMSE** and **MAE** are reported on the log-quantity scale while **MAPE** is computed on the raw quantity scale. The final two columns summarise the per-purpose model's advantage: ΔR^2 is the R^2 difference (positive values favour the per-purpose model) and **Δ RMSE %** is the percentage reduction in RMSE achieved by the per-purpose specification.

Note: Only the significant Δ RMSE's row are retained. The full table was included in the appendix Section 10.3

Product	Segment	R² (PP)	R² (Pool)	RMSE (PP)	RMSE (Pool)	MAPE (PP)	MAPE (Pool)	Δ R²	Δ RMSE%
Cheesecream Latte	Work	0.921	0.668	0.145	0.297	11.5	24.3	+0.252	+51.2%
Caffè Latte	Leisure	0.699	0.780	0.311	0.266	25.5	21.5	–0.081	–16.8%
Caffè Latte	Work	0.932	0.892	0.187	0.235	14.7	18.7	+0.040	+20.7%
Salted Caramel Mac.	Work	0.751	0.458	0.332	0.490	27.1	43.6	+0.293	+32.2%
Pecan Crunch Latte	Leisure	0.623	0.630	0.422	0.419	36.7	36.3	–0.007	–0.9%
Pecan Crunch Latte	Work	0.900	0.641	0.230	0.435	18.6	38.0	+0.259	+47.2%
Vanilla Latte	Leisure	0.698	0.701	0.386	0.383	32.2	31.9	–0.004	–0.6%
Vanilla Latte	Work	0.826	0.703	0.287	0.375	23.6	31.3	+0.123	+23.4%

Table 12: Per-purpose (PP) vs combined model metrics, evaluated on segment-specific subsets. Both models estimated with Optuna-tuned hyperparameters. Log-scale RMSE and MAE; MAPE in percentage points.

5.2.3 Per-Purpose versus combined Model Comparison

In the **Work segment**, the per-purpose model outperforms the combined specification across all ten products without exception. The average **R^2 improvement is 0.150**, ranging from +0.040 for Caffè Latte to +0.293 for Salted Caramel Macchiato, and the average RMSE reduction is approximately 29%. The improvement is largest for products where the combined model struggles most severely: Salted Caramel Macchiato sees its R^2 rise from 0.458 to 0.751, Pecan Crunch Latte from 0.641 to 0.900, and Cheesecream Latte from 0.668 to 0.921. In MAPE terms, the per-purpose model reduces percentage prediction errors by **2 to 19 percentage** points across the Work segment, with the largest MAPE gains appearing in products where the combined model's **MAPE exceeds 35%.** This consistent pattern confirms that the combined model's loss function — dominated by the approximately 92% of observations that come from Leisure stores — allocates insufficient modelling capacity to the structurally distinct Work segment.

In the **Leisure segment**, the picture is more nuanced. The per-purpose model delivers marginal improvements for the majority of products (mean R^2 difference of +0.001), but five of the ten Leisure comparisons show the combined model performing slightly better. Four of these cases are negligible in magnitude: Classic Caramel Macchiato ($\Delta R^2 = -0.007$), Pecan

Crunch Latte (−0.007), Vanilla Latte (−0.004), and Cocoa Latte (−0.001). These differences are small enough to be attributable to tuning variance and are not economically meaningful.

The main exception is **Caffè Latte** in Leisure, where the **combined model outperforms the per-purpose model** ($R^2 = 0.780$ vs. 0.699), a **gap of 0.081** and the **largest reversal** in the comparison. As the highest-volume product, Caffè Latte appears to exhibit relatively homogeneous demand across segments, allowing the combined model to benefit from shared patterns in the full dataset. The weaker per-purpose Leisure result may also reflect **mild overfitting** in hyperparameter selection. This suggests that while the per-purpose approach is generally superior, products with similar cross-segment demand may benefit from pooled estimation.

Overall, however, the evidence still **favours the per-purpose specification**, which is therefore adopted for the subsequent SHAP-based price sensitivity analysis. Its main advantage is a **simpler segment-specific** specification without cross-purpose interaction terms, though the Caffè Latte Leisure results should be interpreted with this trade-off in mind. Despite the fact that splitting the data reduces the Work sample to 1,829 observations, increasing the **risk of overfitting** and preventing the model from borrowing common structure from the Leisure segment. And, that maintaining 20 separate models rather than 10 combined ones increases the burden of retraining, monitoring, and validation. These drawbacks are acknowledged, but are outweighed by the broader gains in predictive accuracy.

5.2.4 Model Interpretation (Shap, PDP/ICE)

The three products presented here—**Cheesecream Latte, Americano, and Hazelnut Latte**—were chosen as representative examples, while the remaining SHAP plots are reported in the Appendix. They also reflect **different positions** in the **popularity ranking**, with Cheesecream Latte being the most popular of the three, followed by Americano, and then Hazelnut Latte. This is useful because many products display broadly similar SHAP structures, but the relative balance between price, lagged demand, and time effects appears to shift as product popularity declines. A short anaylsis on **caffe latte** was also done to explain the finding in the model selection.

5.2.4.1 Cheesecream Latte

Cheesecream Latte is the most **anomalous** product, exhibiting price behavior that defies standard demand assumptions in both segments.

The dominance of t in the Work segment likely reflects a gradual diffusion process: as Cheesecream Latte was introduced or promoted over the observation window, office-based consumers — whose choices are often shaped by workplace norms, peer visibility, and repeated exposure — progressively adopted it as a routine order. In this context, accumulating familiarity over time overrides the role of price, which is further muted by the likelihood that Work-segment purchases are expensed or subsidised, decoupling the individual consumer from direct price sensitivity.

The dominance of short-term lags in the Leisure segment is consistent with habitual but episodic consumption: leisure visitors return at irregular intervals and, when they do, tend to reorder what they most recently enjoyed. This produces a strong autoregressive signal in demand without a sustained upward trend, because purchases are driven by recalled satis-

faction rather than by prolonged exposure or social reinforcement in a shared environment like a workplace.

Together, these patterns suggest that the same product occupies fundamentally different roles across segments — a socially diffusing novelty in Work versus a personally habitual treat in Leisure — which justifies treating the two segments as structurally distinct rather than as a single pooled demand process.

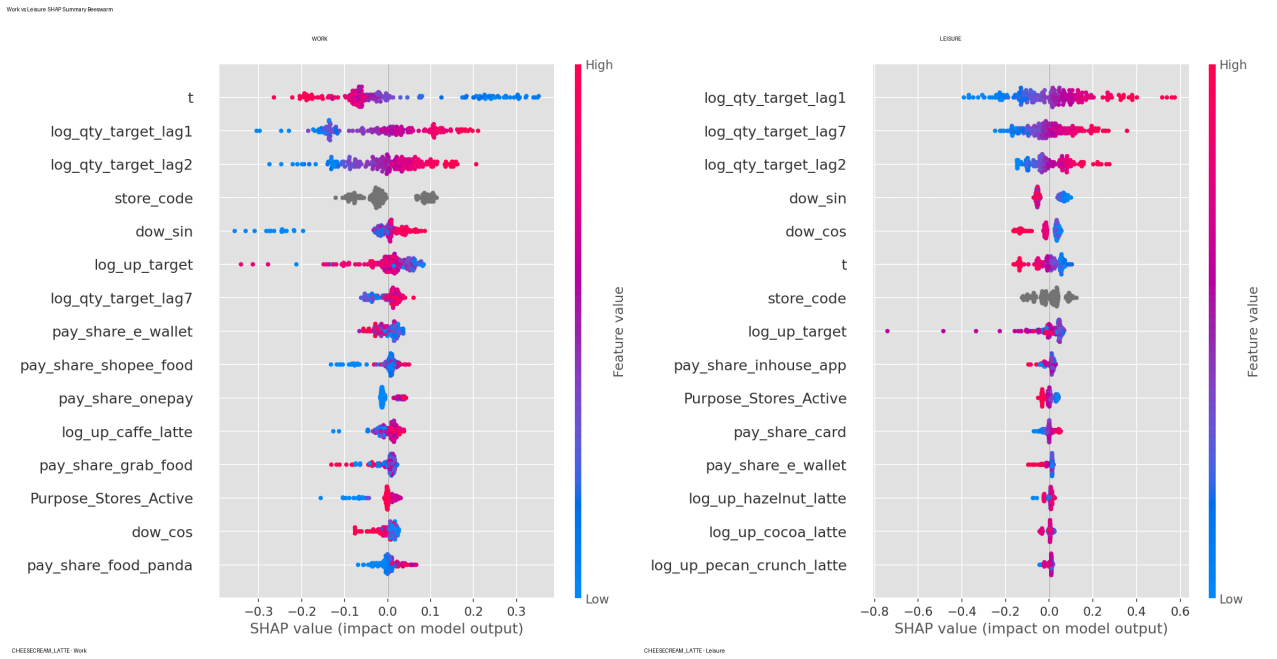


Figure 9: Shap: Cheese Cream Latte, Work vs Leisure

The Work PDP is largely flat with a narrow, sharp trough around price 13.5 that recovers immediately, pointing to a potential localised data artifact rather than a genuine price response, and the ICE lines confirm this: all stores dip and recover together, with otherwise homogeneous and inelastic trajectories. This synchronised behaviour rules out store-specific demand differences as the cause and instead points toward a shared data-level artefact — all stores experienced the same anomalous condition simultaneously, which is more consistent with a calendar-driven event or a data collection issue than with a true price response. This presumably happened because of data sparsity at that point.. If very few observations fall near 13.5 in the training data, the model may have learned an unstable or noisy relationship in that narrow band.

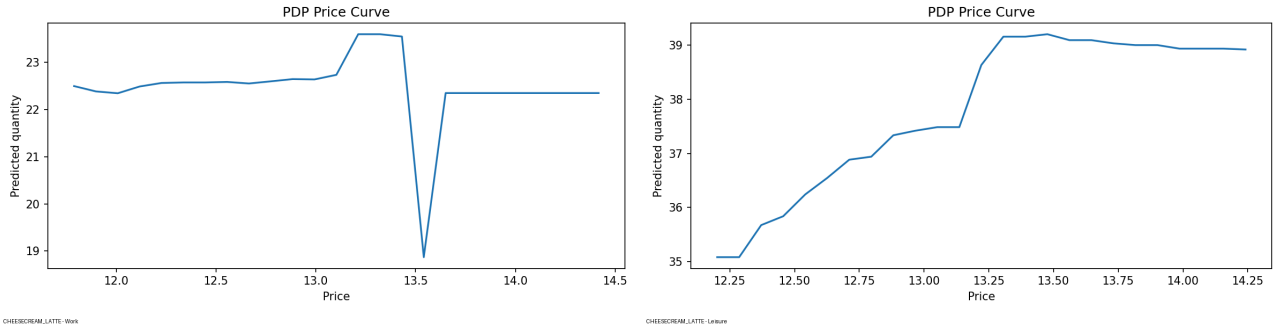


Figure 10: PDP: Cheese Cream Latte, Work vs Leisure

This synchronised behaviour rules out store-specific demand differences as the cause and instead points toward a shared data-level artefact — all stores experienced the same anomalous condition simultaneously, which is more consistent with a calendar-driven event or a data collection issue than with a true price response.

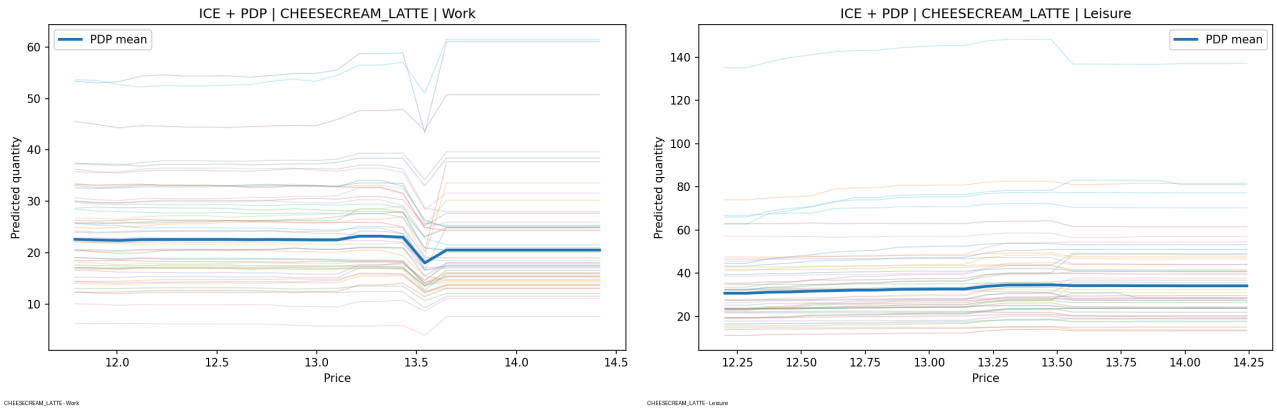


Figure 11: ICE: Cheese Cream Latte, Work vs Leisure

A plausible interpretation is that Cheesecream Latte functions as a **premium or prestige product** in Leisure contexts, where a higher price may signal quality or desirability to discretionary consumers — broadly consistent with Veblen-like behavior. However, the positive relationship may also partly reflect omitted variables correlated with both price and demand, and should therefore be treated cautiously in any causal framing. Regardless of mechanism, given that all the observations in the ICE plot demonstrate the same pattern, it can be concluded that the curve observed is not a typical downward slopping demand curve.

5.2.4.2 Americano

Americano presents the sharpest Work-versus-Leisure contrast among all products analyzed. In Work stores, `store_code` dominates the SHAP ranking while `log_up_target` falls to approximately 16th place, far below lagged sales, day-of-week, and payment variables. The PDP curve is nearly flat across the lower-to-mid price range before declining steeply at higher prices, and the ICE lines are strikingly parallel and horizontal — together providing strong evidence that Americano demand in Work contexts is governed by store identity and habitual purchasing patterns rather than price. In Leisure, own price rises to second place in the SHAP ranking, but the color pattern is asymmetric and non-monotonic: blue low-price observations cluster in the center while purple points appear on both the positive and negative SHAP sides, indicating heterogeneous rather than uniformly negative price effects. The Leisure PDP is highly turbulent with a sharp spike near price 9, and the ICE lines diverge substantially, confirming that price effects in this segment are real but context-dependent, mediated by interactions with store type, lagged demand, and occasion-driven behavior.

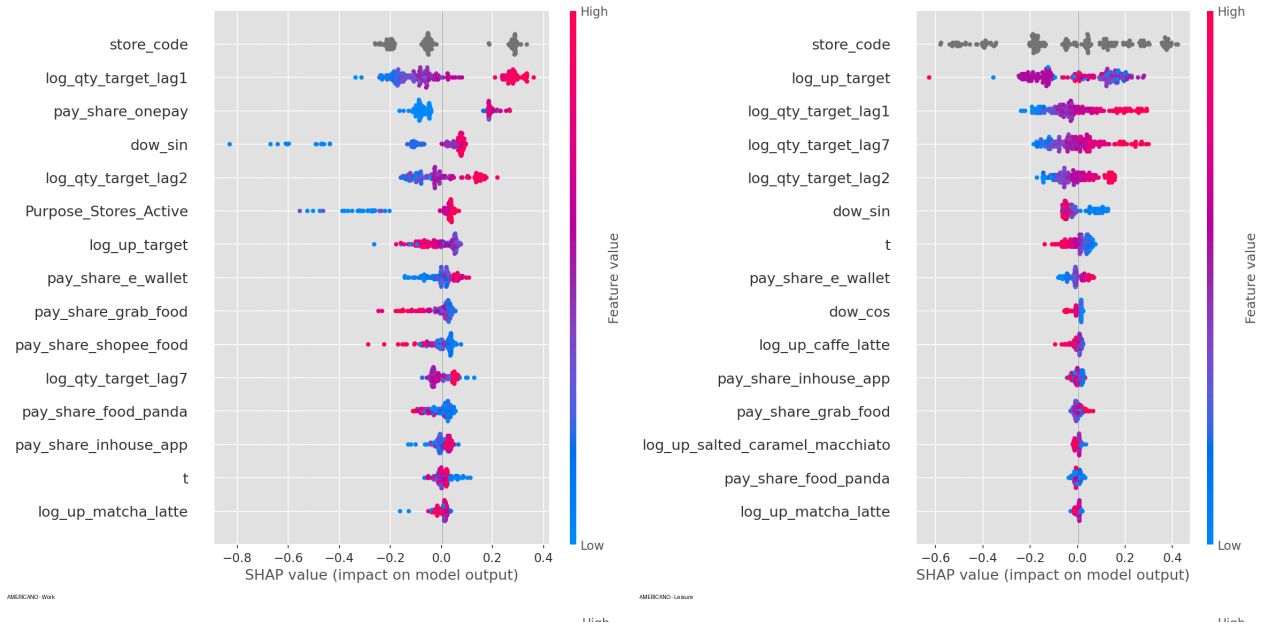


Figure 12: Shap: Americano, Work vs Leisure

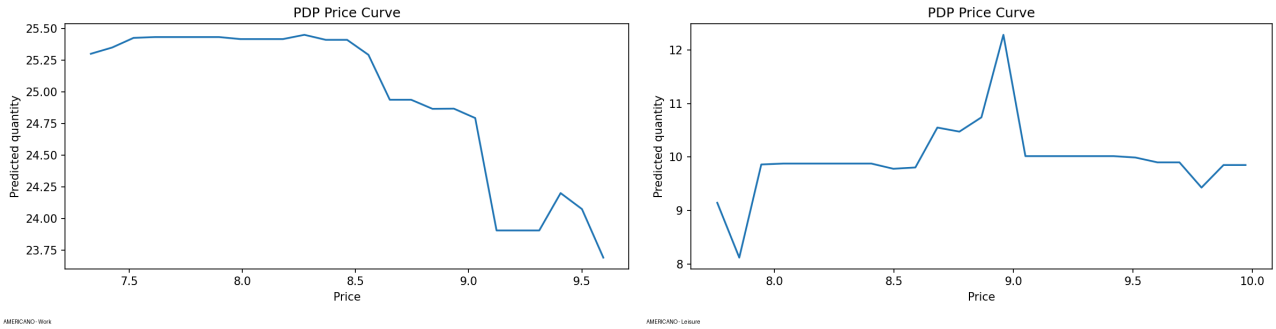


Figure 13: PDP: Americano, Work vs Leisure

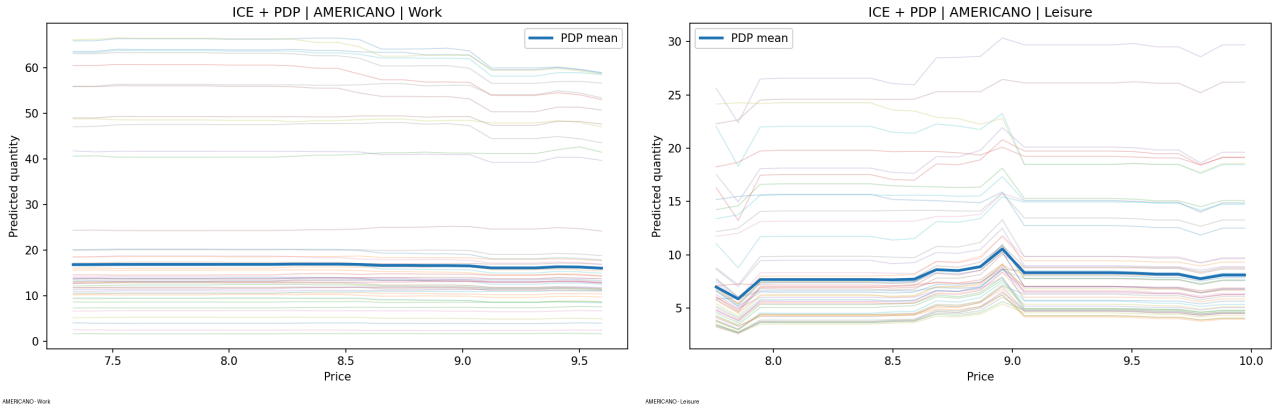


Figure 14: ICE: Americano, Work vs Leisure

5.2.4.3 Hazelnut Latte

As for Hazelnut Latte, it emerges as the most **conventionally price-sensitive** product in the portfolio. In both Work and Leisure segments, `log_up_target` ranks as the single **most important** SHAP predictor, placing own price above lagged sales, store identity, and payment-channel variables. This features were only available for the products with **lower popularity**. The SHAP colour pattern shows that higher feature values (higher prices) tend to be associated with negative SHAP contributions in both segments, consistent with conventional downward-sloping demand, though the relationship is not perfectly monotonic —

purple and mixed observations appear throughout, reinforcing that the price effect interacts with store context and timing rather than operating uniformly.

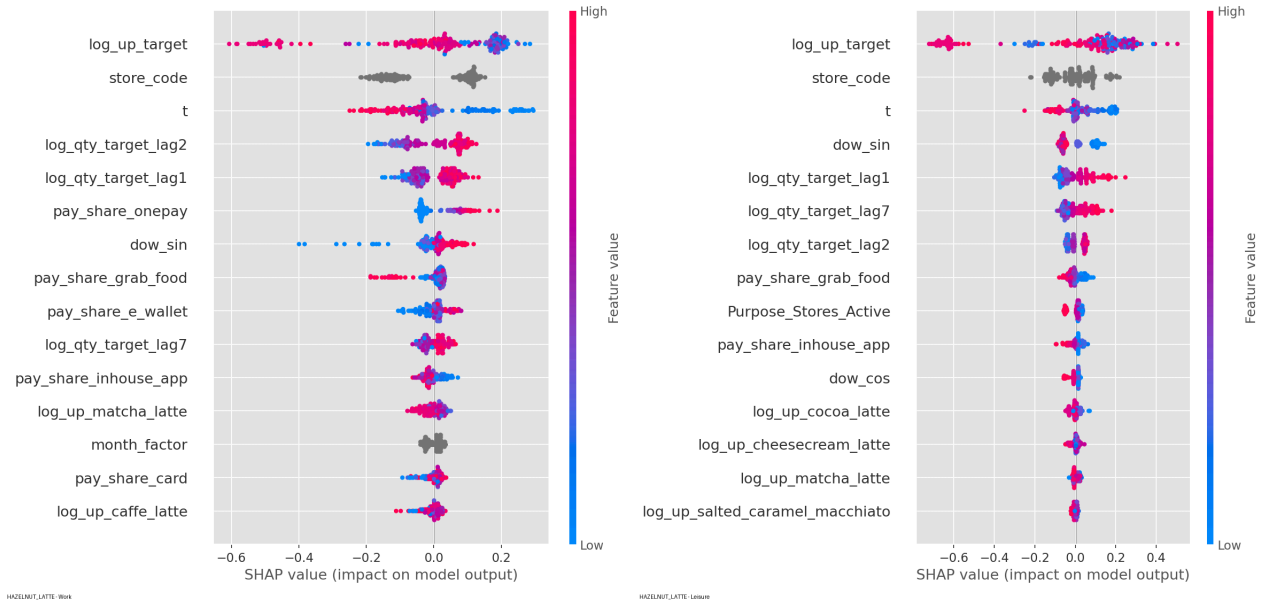


Figure 15: Shap: Hazelnut Latte, Work vs Leisure

The Work PDP for Hazelnut Latte is smooth and threshold-shaped: demand holds relatively stable through mid-range prices before declining sharply as price approaches the upper end of the observed range. This pattern suggests the existence of a price ceiling below which demand is resilient, and above which consumers substitute away. The ICE lines in the Work segment are largely parallel and co-moving, indicating a homogeneous price response — the decline at higher prices is shared broadly across stores rather than being driven by a subset of particularly sensitive outlets.

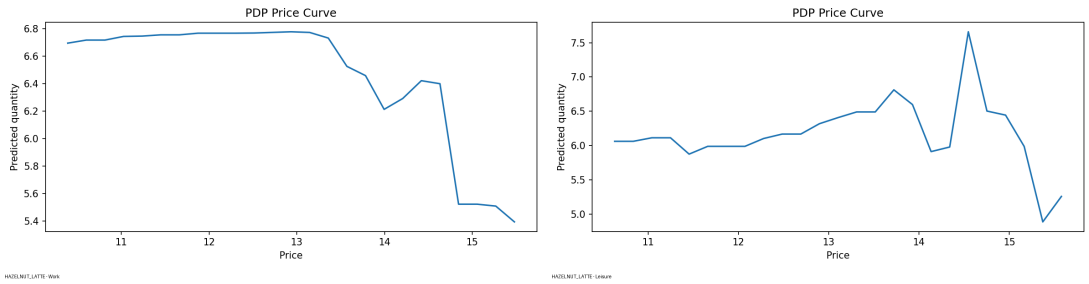


Figure 16: PDP: Hazelnut Latte, Work vs Leisure

In Leisure stores, the PDP becomes more **volatile and non-monotonic**, with a visible hump in the mid-to-high price region. This hump is driven by subgroup heterogeneity rather than a genuine aggregate reversal: the **ICE plots** show that Leisure store trajectories **diverge and cross**, particularly at higher price points, meaning that some stores see demand increase in a price band where others see it fall. The net PDP averages these opposing effects, producing the non-monotonic shape. This **contrast** between the two segments is **critical for pricing strategy**: while Work stores can be treated with a single, relatively predictable demand curve, Leisure stores require a more granular, store-level approach that accounts for local demand conditions and competitive context.

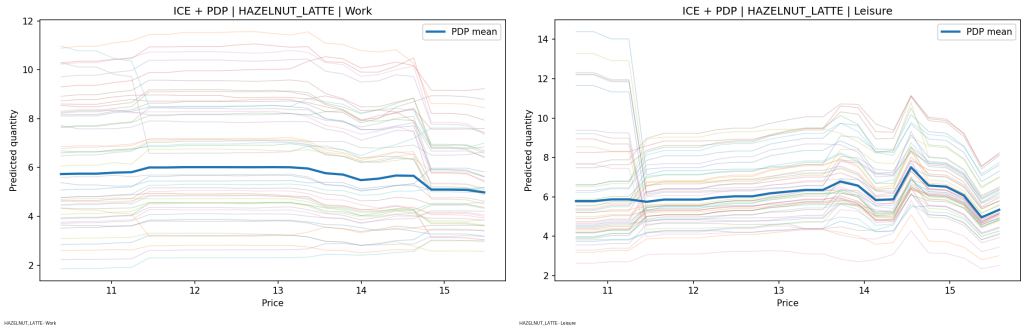


Figure 17: PDP: Hazelnut Latte, Work vs Leisure

5.2.4.4 Overall Discussion

A consistent finding across all three products is the importance of `store_code` as a top-ranked SHAP predictor. This indicates that baseline demand conditions vary substantially across stores, even after controlling for the Work–Leisure segmentation. Because `store_code` is included as a **categorical feature**, its SHAP **colour** gradient **should not be interpreted numerically**; rather, it shows that the model assigns systematically different baseline demand levels to different outlets. In effect, `store_code` acts as a proxy for unobserved store-level effects — location quality, foot traffic, local competition, and customer demographics — confirming that **demand is heterogeneous** across outlets even within the same product and purpose segment.

A further observation concerns **cross-product substitution** effects. Although rival product prices were included as features and contributed to tree-splitting decisions, the SHAP dependence plots for most products show values **tightly clustered around zero**, indicating that rival prices **do not exert a prominent substitution effect** on the target product's demand. A small number of exceptions — notably the rival prices of Hazelnut Latte in the Cheesecream Latte and Cocoa Latte models — display a monotonic SHAP pattern, but in a direction **opposite** to standard substitution theory: a decrease in the rival's price is associated with higher predicted demand for the target product, consistent with **complementarity rather than substitution**. These patterns should be interpreted with caution, as SHAP values reflect correlations learned by the model rather than causal mechanisms; co-movement in prices during promotional events or seasonal demand shifts may produce associations that would not materialise under an isolated price intervention.

Taken together, the three products illustrate a spectrum of price sensitivity regimes. **Hazelnut Latte** conforms most closely to textbook demand theory, with own price as the dominant predictor and a **clear negative price–quantity relationship**, especially in Work stores. **Americano** sits in the middle: **price is largely irrelevant** in Work contexts where habitual consumption dominates, but becomes a meaningful and heterogeneous driver in Leisure stores. **Cheesecream Latte** is the **outlier**, where price not only fails to suppress demand but appears to boost it in Leisure settings, consistent with premium or prestige positioning. These findings underscore the need for product- and segment-specific pricing strategies rather than uniform elasticity assumptions across the portfolio.

Additionally, if we look into **cafe latte** to explain the similar r2 in pooled model before, both Work and Leisure PDP curves actually share a similar shape: a relatively flat plateau at lower prices, followed by a decline as price increases. The overall demand pattern looks

roughly the same across both segments. This is exactly why the combined model outperforms the per-purpose model for Caffè Latte Leisure. Since the demand patterns look roughly similar across segments, the combined model benefits from pooling all the data to learn that shared structure more robustly.

However, for Work, the decline starts around RM 11 and falls sharply (losing 6 units from RM 11 to RM 13). For Leisure, the drop begins later around RM 11.5 and is more gradual (2–3 units lost over a similar range). Because the curves share the same general pattern, the revenue-maximizing price just lands at different points along that curve. Work hits the point where volume loss outweighs per-unit gain at RM 11.26, while Leisure's gentler slope means you can push to RM 13.41 before that trade-off flips. (Which suggest the big difference in the optimal price as inlcuded in the revenue optimisation)

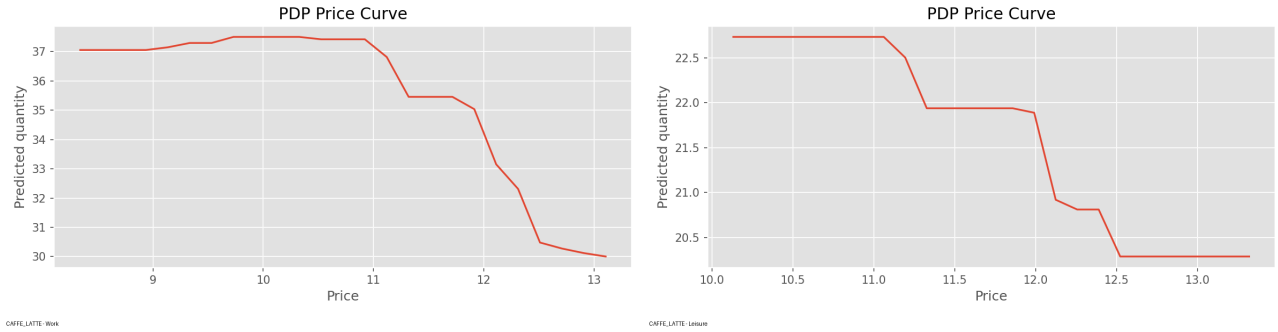


Figure 18: PDP: Hazelnut Latte, Work vs Leisure

Plots for the remaining products were included in the appendix Section 10.9

5.2.5 Revenue Optimisation

5.2.5.1 Results

Table 13 reports the revenue-maximising price for each product–purpose combination, along with the associated mean predicted daily quantity per store at the optimal price.j

Product	Leisure Price (RM)	Work Price (RM)	Δ Price (RM)	Δ Price (%)
Cheesecream Latte	14.35	13.46	–0.89	–6.2%
Caffè Latte	13.41	11.26	–2.15	–16.0%
Americano	8.98	9.02	+0.04	+0.4%
Matcha Latte	13.42	13.48	+0.07	+0.5%
Classic Caramel Mac.	14.59	14.56	–0.03	–0.2%
Salted Caramel Mac.	16.91	15.63	–1.29	–7.6%
Hazelnut Latte	14.61	14.54	–0.06	–0.4%
Pecan Crunch Latte	14.65	13.36	–1.29	–8.8%
Vanilla Latte	14.59	14.58	–0.01	–0.1%
Cocoa Latte	13.48	13.49	+0.00	+0.0%

Table 13: Revenue-maximising prices by product and segment.
Δ Price = Work price minus Leisure price. Prices in Malaysian Ringgit (RM).

The results reveal **three** distinct pricing patterns across the portfolio.

First, the majority of products (seven of ten) show revenue-maximising prices that are **nearly identical** across segments, with Work–Leisure differences of less than RM 0.10. For

these products — Americano, Matcha Latte, Classic Caramel Macchiato, Hazelnut Latte, Vanilla Latte, and Cocoa Latte — the **optimal price point** is effectively **segment-invariant**, suggesting that current uniform pricing across store types is approximately **revenue-optimal**.

Second, three products exhibit **materially lower revenue**-maximising prices in Work stores: **Caffè Latte** (RM 11.26 vs 13.41, a 16% differential), **Salted Caramel Macchiato** (RM 15.63 vs 16.91, 7.6% lower), and **Pecan Crunch Latte** (RM 13.36 vs 14.65, 8.8% lower). For **Caffè Latte**, the Work segment's substantially lower optimal price reflects the product's role as a routine, **habitual purchase** in office environments — where demand volume is high (mean predicted quantity of 32.6 cups per store-day, the highest in the Work segment) but price sensitivity is **elevated** because the purchase decision is made daily and competes directly with lower-cost alternatives such as Americano.

Third, Cheesecream Latte warrants separate discussion. Its Leisure optimal price of RM 14.35 lies at the **upper bound** of the observed PDP price range. Given that the PDP is **monotonically increasing** over this range, the model suggests that higher prices continue to raise predicted demand; however, this also implies that the current price variation may be insufficient to identify a true optimum without extrapolation beyond observed data. In Work stores, the slightly lower optimal price of RM 13.46 may reflect the artefactual dip previously observed in the PDP, rather than a stable underlying demand relationship.

5.3 Comparison of Results

Results from both CatBoost and Log-Log linear models both indicate that the effect of rival goods on a target product is generally less significant: the linear models only maintained at most two rival products that are both statistically and economically significant, and the SHAP values from Catboost infer likewise. Furthermore, both CatBoost and the Linear Model appear to show differences in demand behaviour between segments, although formal statistical tests are necessary to conclude on its statistical significance.

However, for the Leisure Segment, by looking at the PDP curve of Cheesecream Latte from CatBoost and comparing it to the own-price elasticity of Cheesecream Latte from the Linear Model, it can be observed that they both provide conflicting results. While the PDP curve suggests that demand increases as price increases, the own-price elasticity demonstrates otherwise. But since no measures were taken for CatBoost to account for endogeneity, the results from the PDP should not be used for causal inference. As such, this discrepancy can be explained as the lack of ability of the CatBoost model to isolate the causal effects of price elasticity due to the nature of its design and purpose.

6 Limitations & Future Recommendations

Table 14 and Table 15 summarises the key limitations of the modelling and optimisation framework, together with their practical implications and potential mitigations.

6.1 Log-Log Linear Demand Models using 2SLS Regression

Limitation	Implication	Mitigation / Future Work
OLS estimates may be biased due to endogeneity. IV/2SLS methods rely on the exclusion restriction, which is dif-	Price elasticity estimates may not achieve perfect causal identification, limiting theoretical precision.	Adopt a decision-oriented perspective where the value of estimates is judged by their ability to inform actionable,

Limitation	Implication	Mitigation / Future Work
ficult to verify and often idealized in practice.		data-driven pricing decisions rather than by strict causal identification.
Work and leisure segments are modelled separately rather than in a single interacted specification, reducing the effective sample size per regression.	Lower estimation efficiency and higher coefficient variance, particularly for the smaller work segment.	Collect larger segment-level samples to improve precision. Alternatively, explore pooled models with interaction terms once sufficient work-segment observations are available.
2SLS estimates with robust standard errors are not fully efficient under heteroskedasticity and autocorrelation.	Parameter estimates remain consistent but standard errors may be larger than necessary, reducing statistical power and widening confidence intervals.	Employ Generalised Method of Moments (GMM) or feasible GLS-IV estimators that explicitly model the error structure, yielding efficiency gains while preserving consistency.
Individual Log-log linear models that are modeled individually do not explicitly capture substitution patterns between products.	Estimated cross-price effects may not fully reflect underlying consumer choice behaviour; restrictive economic assumptions are imposed.	Implement structural demand systems such as the Almost Ideal Demand System (AIDS) for smaller product sets, or the Berry–Levinsohn–Pakes (BLP) model for larger assortments, to incorporate theory-consistent substitution patterns.
Hausman IV was utilized, but the implication of having delivery app services was not considered	Delivery app can have region-wide promotion. Under such circumstance, the Hausman IV will fail to isolate price changes due to cost movements	Select additional IVs and utilize and over-identified approach

Table 14: Summary of key limitations, their implications, and recommended directions for future research.

6.2 Catboost Models

Limitation	Implication	Mitigation / Future Work
Endogeneity in price–demand relationship	The model is estimated from observational data without randomised pricing experiments. Reverse causality or omitted-variable confounding may cause counterfactual price simulations to understate (or overstate) the true demand response, biasing optimal price recommendations.	Rich controls (lagged sales, store identity, day-of-week, payment channels, rival prices) mitigate but do not eliminate the concern. Results should be treated as indicative rather than causal. Validation through controlled A/B pricing experiments is recommended.
Static optimisation	The framework identifies a single best price per product–purpose segment, averaged over all store-days. It does not capture dynamic effects such as reference-price formation, inventory depletion, or competitive retaliation that may alter demand in subsequent periods.	Extending the framework to a dynamic pricing model that accounts for intertemporal demand feedback and inventory constraints.
Single-product optimisation	Each product is optimised independently while holding rival prices at their observed levels. Cross-product substitution is only partially captured: raising one product's price may shift demand to substitutes, affecting total portfolio revenue in ways the single-product optimiser ignores.	Joint multi-product price optimisation across the full beverage portfolio (Ferreira, Lee, and Simchi-Levi, 2016).
PDP and ICE reflect model behaviour, not causal effects	PDP and ICE curves show how the fitted model's predictions change as one feature varies, holding others fixed. They are not causal: when features are correlated or the model captures spurious patterns, the implied "all else equal" interpretation may not reflect the effect of a real intervention.	Interpret PDP/ICE as diagnostic tools for understanding model behaviour rather than as estimates of causal demand elasticities. Complement with SHAP dependence plots and domain-validated experimental evidence where possible.

Limitation	Implication	Mitigation / Future Work
Interpolation-constrained price optimisation	CatBoost cannot extrapolate beyond observed price ranges; out-of-range inputs simply return the nearest leaf value. Reported optimal prices are therefore within-sample — if the true optimum lies outside historical variation, the model cannot discover it.	The log-log model complements CatBoost by enabling extrapolation beyond observed prices, at the cost of stronger functional form assumptions. Controlled pricing experiments would expand the observed range and reduce this constraint.

Table 15: Summary of key limitations of the CatBoost demand modelling and price optimisation framework.

7 Conclusion

This study evaluates demand for a coffeehouse chain in Selangor using both econometric (Log-Log linear models estimated via 2SLS) and machine learning (CatBoost gradient boosting) methods, applied separately to Work and Leisure consumer segments.

The findings from the 2SLS regression method suggests that the coffeehouse chain has the potential for market dominance, as all ten products exhibit negative and inelastic demand (> -1) across both segments. This inelasticity indicates the possibility of high profitability, especially if the chain holds a dominant position in the market. The "next steps" for businesses that dominate the market may include focusing on customer retention and exploring opportunities to expand into new geographic markets or consumer segments, where the brand's pricing power may remain strong. At the same time, results from 2SLS regression also highlights that inelastic demand does not justify across-the-board price increases. For example, while Cheesecream Latte shows relatively inelastic demand (-0.435) in the Leisure segment, raising its price is not advisable due to its complementarity with other menu items. A price hike could reduce the demand for other products, potentially lowering overall sales and revenue, despite the possible increase in Cheesecream Latte's own revenue. Therefore, pricing decisions must consider product interdependencies and should not be made in isolation, especially for bestsellers that drive customer visits.

The CatBoost model provides additional insights into consumer behavior, revealing that Work consumers tend to exhibit stable, predictable demand patterns driven by routine, while Leisure consumers display more dynamic and context-dependent responses. This difference is significant for pricing strategies, as Work consumers' purchasing habits are more predictable, while Leisure consumers show greater price sensitivity and variability. The analysis confirms that separate models for each segment outperform a combined model, especially in the Work segment. Finally, the revenue optimization simulation suggests that segment-differentiated pricing should be considered only for certain products. Specifically, for three items—Caffè Latte, Salted Caramel Macchiato, and Pecan Crunch Latte—the optimal price varies between the Work and Leisure segments, with Work prices being 7–16% lower. For the other seven products, uniform pricing appears close to optimal. This selective segmentation strategy simplifies implementation and reduces managerial complexity.

In conclusion, combining the causal insights from the 2SLS model with the predictive flexibility of CatBoost provides a more comprehensive approach to pricing analytics. The joint use of both models leads to more robust and actionable pricing recommendations, offering segment-specific strategies that optimize profitability and ensure a nuanced understanding of consumer demand.

8 Appendix: Dataset Details

8.1 Data Cleaning Details

Holiday.Name	S0003	S0007	S0009	S0015	S0019	S0026	S0029	S0040	S0047	S0057	S0058	S0062	S0065	S0083	S0084	Total
Public Holiday										1	1					2
Awal Muharram										3	3				2	8
Birthday of SPB Yang di Pertuan Agong							1			2	2				1	6
Birthday of The Sultan of Selangor										2	2				1	5
Chinese New Year										1	1				1	3
Chinese New Year Holiday										2	2				2	6
Christmas Day										2	2				1	5
Deepavali										2	2				1	5
Deepavali Holiday										1	1					2
Father's Day										2	2				1	5
Hari Raya Aidilfitri	3						2			3	2				2	12
Hari Raya Aidilfitri Holiday	1									1						2
Hari Raya Haji										1	2				1	4
Labour Day										2	2				1	5
Malaysia Day									1	1	1					3
Maulidur Rasul									1	2	2			1	1	7
Mother's Day										2	2				1	5
National Day									1	2	2			1	1	7
New Year's Day										1	1					2
No_Holiday		9	3	1	1	2	43	1	42	143	145	7	3	23	86	509
Ramadan		14					14		1	8	9				9	55
Thaipusam							1			1	1				1	4
Wesak Day										2	2				1	5
Total	4	23	3	1	1	2	61	1	46	187	189	7	3	25	114	667

Figure 19: Missing Date Table

Holiday.Name	S0003	S0007	S0009	S0029	S0040	S0047	S0057	S0058	S0062	S0065	S0083	S0084	Total
Public Holiday							1	1					2
Awal Muharram							1	1				1	3
Birthday of SPB Yang di Pertuan Agong				1			2	2				1	6
Birthday of The Sultan of Selangor							1	1					2
Chinese New Year Holiday							1	1				1	3
Christmas Day							1	1					2
Deepavali							1	1				1	3
Deepavali Holiday							1	1					2
Hari Raya Aidilfitri	1			1			1	1				1	5
Hari Raya Haji							1	2				1	4
Labour Day							1	1					2
Maulidur Rasul						1	2	2			1	1	7
National Day						1	1	1					3
New Year's Day							1	1					2
No_Holiday		4	2	23	1	24		3	3	2			62
Ramadan		8		8		1		1				1	19
Thaipusam				1			1	1				1	4
Wesak Day							1	1					2
Total	1	12	2	34	1	27	18	23	3	2	1	9	133

Figure 20: Missing Date Table(Weekday)

Holiday.Name	S0003	S0007	S0015	S0019	S0026	S0029	S0047	S0057	S0058	S0062	S0065	S0083	S0084	Total
Awal Muharram								1	1				1	3
Chinese New Year								1	1				1	3
Chinese New Year Holiday								1	1				1	3
Deepavali								1	1					2
Father's Day								2	2				1	5
Hari Raya Aidilfitri	1							1						2
Hari Raya Aidilfitri Holiday	1							1						2
Malaysia Day							1	1	1					3
Mother's Day								2	2				1	5
National Day								1	1			1	1	4
No_Holiday		4	1	1	2	14	11	143	141	3	1	23	86	430
Ramadan		4				4		8	8				8	32
Total	2	8	1	1	2	18	12	163	159	3	1	24	100	494

Figure 21: Missing Date Table(Weekend)

Cleaning Step	Action Taken	Rationale / Statistical Impact
Store Inclusion	Removed 6 Stores	Exclusion of outlets with < 6 months of data to ensure robust seasonality estimates.
Regional Scoping	Removed KL	Restricted analysis to the Selangor market to maintain geographic consistency.
Soft Launch Adjustment	Adjusted 126 Rows	Moved sales dates before opening to Opening.Date -1 to retain "soft launch" volume without biasing trends.
Naming Correction	Standardized Name	Merged 'DARK CHOCOLATE LATTE' into 'DARK CHOCO LATTE' to consolidate historical product performance.
Anomaly Removal	Removed 2 Rows	Purged negative pricing entries identified as system errors or data entry anomalies.
Missing Values	Imputed 'Others'	Combined 'NA' entries with the 'Others' category in Payment_Method to prevent row loss and preserve transaction volume.
Redundancy Reduction	Dropped 2 Columns	Removed 'Sub Category' and 'Product Code' from the modeling phase to minimize multi-collinearity.
Geographic Mapping	Added District/City	Mapped 40 cities into 6 Districts (DOSM) and 3 Economic Groups for spatial structure.
Log-Log Eligibility	Removed 943 Rows	Excluded pricing $l = 0$ to satisfy logarithmic requirements ($\log(P)$) and ensure valid constant price elasticity modeling.

Table 16: Summary of Data Cleaning Procedures.

8.2 Data Description Details

Feature Name	Type	Levels	Description
ID	Integer	NA	Unique identifier for each transaction
Sales.Date	Date	NA	Date of sales transaction
Store.Code	Factor	40	Unique identifier for each store
Opening.Date	Date	NA	Date when the store officially opened
City	Factor	20	City where the store is located
State	Factor	1	State where the store is located
Type	Factor	3	Store type
Venue	Factor	4	Venue category of the store
Payment.Method	Factor	15	Original payment method used
Product.Code	Factor	≈ 200	A unique SKU identifier representing a specific combination of Product Name, Bean, Milk, and Temperature.
Product.Name	Factor	38	Name of product sold
Temperature	Factor	2	Product temperature (Hot, Cold)
Bean	Binary	2	Binary flags for bean origin: Sumatra, Huila or NA.
Milk	Binary	4	Binary flags for milk variety: Fresh, Oat, Soy, Coconut or NA.
Sub.Category	Factor	11	Product sub category
Quantity	Integer	NA	Number of units sold in a transaction
Selling.Price	Double	NA	Listed selling price before discount
Discount	Double	NA	Discount applied to the transaction
Net.Selling.Price	Double	NA	Final selling price after discount
Holiday.Type	Factor	3	Type of holiday

Table 17: Summary of Primary Dataset Variables.

8.3 Feature Engineering Details

Feature	Type	Statistical / Practical Rationale
Time Index (t)	Fixed	Captures deterministic long-term growth or decay trends over the two-year period.
Unit Price	Fixed	Derived from net selling price to estimate price elasticity of demand for core products.
City.Group	Fixed	Clustered by economic profile ('Core Metropolitan', 'Commuter Belt' and 'Outer Growth Periphery') to identify urban demand patterns.
Weekly Cycle	Cyclic Encoding	Encoded as $\sin\left(\frac{2\pi \cdot \text{dow}}{7}\right)$ and $\cos\left(\frac{2\pi \cdot \text{dow}}{7}\right)$. Treats the week as a continuous circle to preserve the temporal continuity between the conclusion of a weekly cycle and the commencement of the subsequent period.
Holiday Dummies	Binary	Filtered for Selangor-specific events to identify structural breaks during festive periods.
Discount Rates	Fixed	Aggregated by product per day (dr_j); captures the depth of promotions on sales volume.
Product Quantities	Fixed	Daily summation of specific product units (Qty_j) to align with store-level modeling.
pay_share_*	Proportion	Share of transactions per payment method to capture shifting consumer preferences(e.g., pay_share_card).
bean_share_*	Proportion	Daily share of specific bean origins to identify flavor-profile drivers (e.g., bean_share_huila).
milk_share_*	Proportion	Daily share of milk varieties used to account for dietary trends (e.g., milk_share_oat).

Table 18: Summary of Feature Engineering and Variable Selection.

8.4 Data Reshaping (Pivoting)

To transition from raw transactional records to a balanced panel dataset suitable for Fixed Effects (FE) estimation, the data underwent a multi-stage transformation pipeline. This process reshaped the data from a "long" transactional format to a "wide" modeling format, allowing for the simultaneous analysis of own-price and cross-price elasticities at the store level.

The initial dataset consisted of approximately 1.9 million individual transaction lines. To establish a consistent unit of observation for time-series analysis, the data was vertically aggregated by Store_Code and Sales_Date.During this stage, three primary metrics were calculated for each product-store-day combination:

- Quantity: The arithmetic sum of units sold, serving as the dependent variable.
- Unit Price: Derived by dividing total net revenue by total quantity to account for intra-day price fluctuations and varying discount applications.
- Discount Rate: The ratio of total discounts to the total gross selling price, representing the average promotional depth for that specific day.

While the aggregated data was structured by store and date, it remained in a "long" format where each product occupied a separate row. This structure is computationally incompatible with estimating cross-price effects, as the long format prevents the model from

associating the price of a substitute (e.g., Caffe Latte) with the quantity of the target product (e.g., Americano) within a single observation row.

To resolve this, a horizontal pivot was executed using the This expanded the dimensionality of the dataset by transforming product names into distinct column headers. To ensure statistical clarity and prevent column name collisions, a prefix-based naming convention was adopted:

- Qty_[Product]: Captures the daily demand volume.
- up_[Product]: Stores the unit price for elasticity testing.

The following tables illustrate the transition from raw transactional records to the analytical wide format required for the fixed-effects model.

Sales.Date	Store.Code	Product.Name	Unit.Price	Quantity
01-01-2024	S001	AMERICANO	10.00	1
01-01-2024	S001	CAFFE_LATTE	12.00	2
02-01-2024	S001	AMERICANO	10.00	1

Table 19: Transactional (Long) Format prior to aggregation.

Sales.Date	Store.Code	Qty_AMERI	up_AMERI	Qty_CAFFE	up_CAFFE
01-01-2024	S001	1	10.00	1	12
02-01-2024	S001	1	10.00	2	12

Table 20: Engineered (Wide) Format used for Fixed Effects modeling.

Table 21 summarises the sample construction process. After removing stores with fewer than 180 days of data and excluding zero or negative price observations, the dataset was aggregated into a daily store-product panel and then split into Work and Leisure segments. The final wide store-date panel contains 22,107 observations.

Step	Observations
Raw transaction data	1,958,127
After removing stores with < 180 days	1,587,587
After removing zero/negative price observations	1,586,644
Daily store–product panel (aggregated)	
Work segment	39,688
Leisure segment	468,950
Wide store–date panel (final sample)	
Work segment	1,829
Leisure segment	20,278

Table 21: Sample construction and aggregation flow of the dataset.

8.5 Univariate Distribution of Products

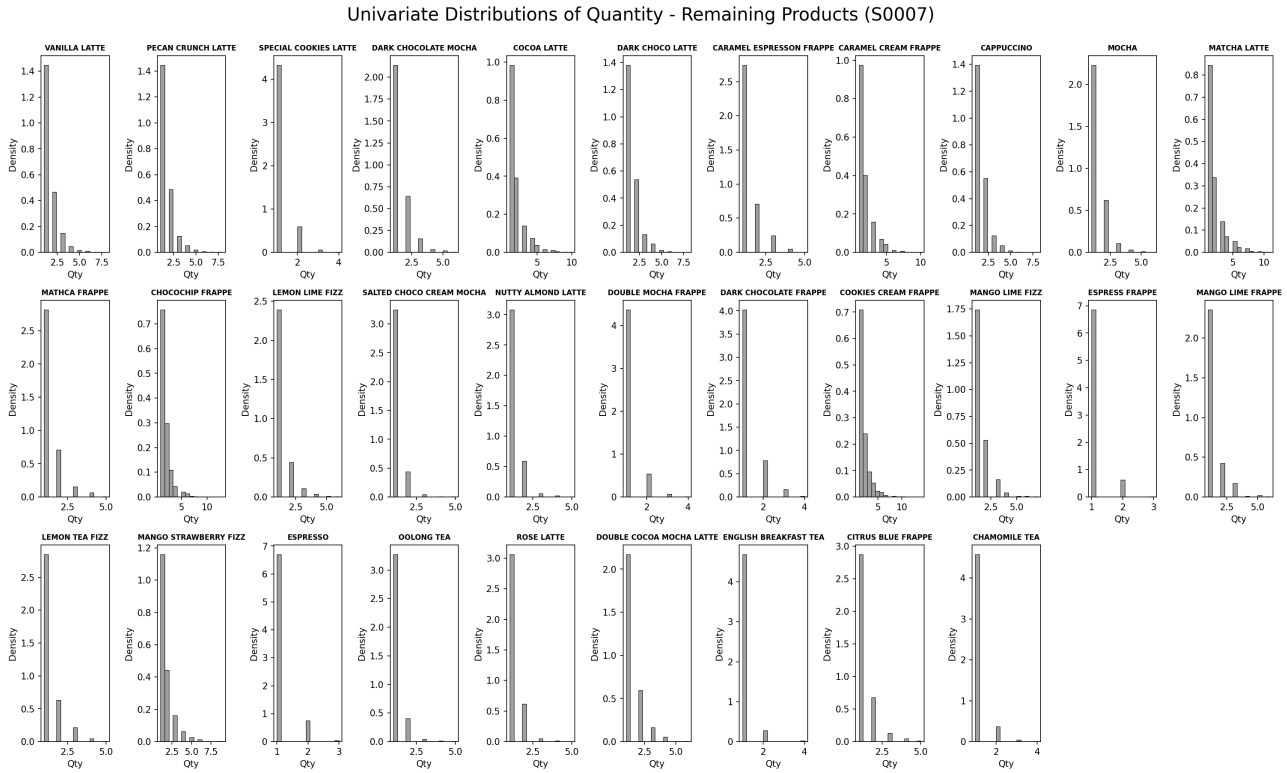


Figure 22: Weekly Sales Volume by Venue.

8.6 ACF and PACF

ACF and PACF plots for other products were also plotted as shown in Section 8.6 below for store S0007.

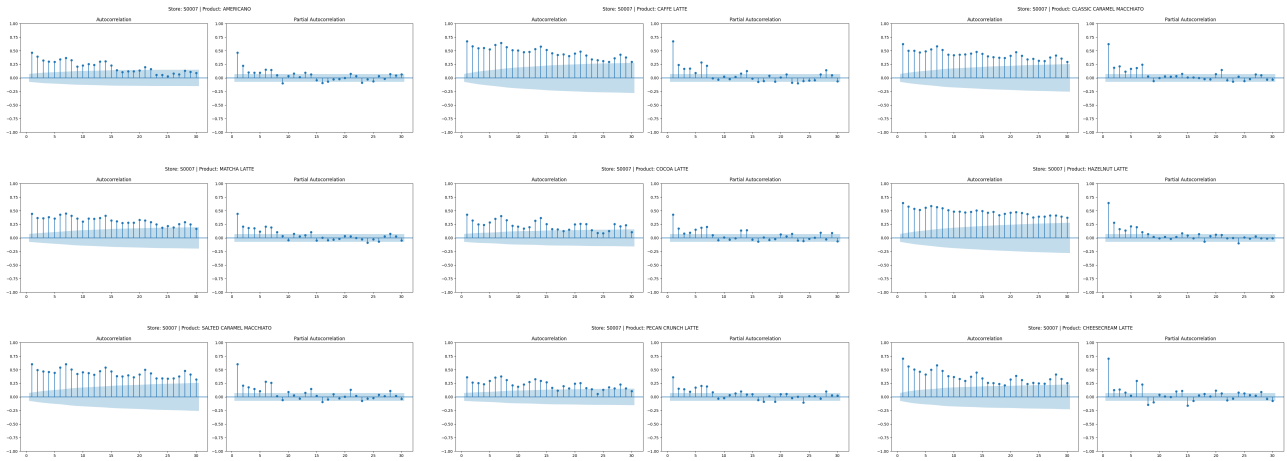
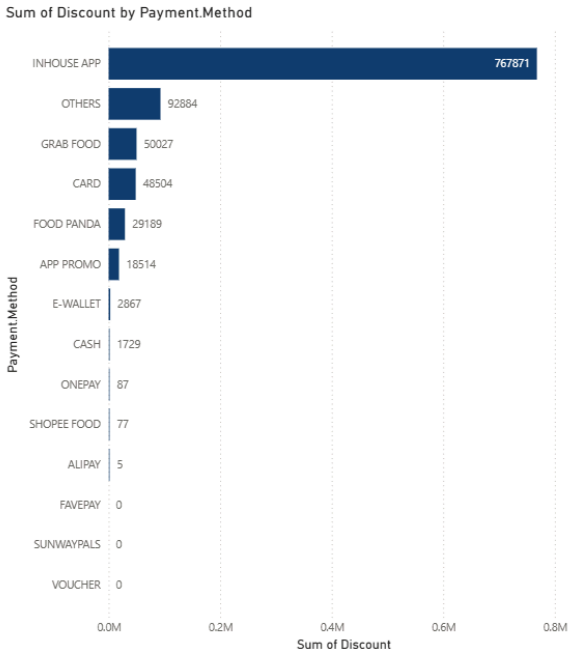
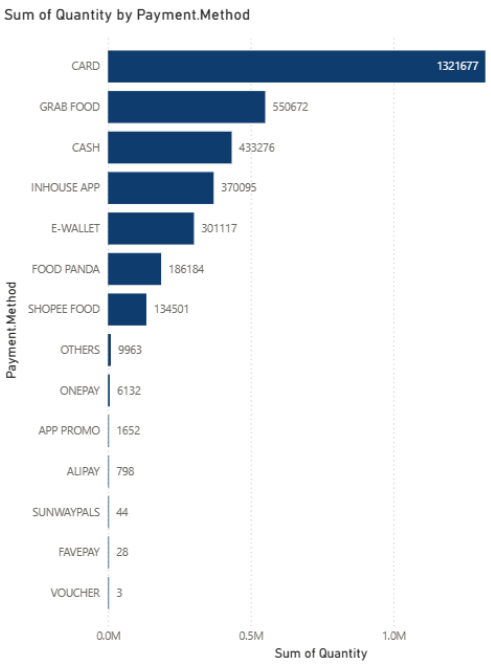


Figure 23: ACF and PACF Diagnostics for Top 9 Products in Store S0007.

8.7 Relationship of Payment Menthod to Sales



(a) Aggregate Discount by Payment Method



(b) Total Quantity by Payment Method

Figure 24: Comparison of Discount Allocation and Transaction Volume across Payment Methods.

8.8 Box-Cox Diagnostic

The table below summarizes the Box-Cox transformation results used to determine the optimal functional form for each product.

Product	Best_λ	CI Lower	CI Upper	Suggested Model
AMERICANO	0.101	0.101	0.101	Log-Log
CAFFE_LATTE	0.2626	0.2626	0.2626	Other
CAPPUCCINO	−0.101	−0.101	−0.101	Log-Log
CARAMEL_CREAM_FRAPPE	−0.2626	−0.2626	−0.2222	Other
CARAMEL_ESPRESSON_FRAPPE	−0.3838	−0.4242	−0.3838	Other
CHAMOMILE_TEA	−2.0	−2.0	−2.0	Other
CHEESECREAM_LATTE	0.1818	0.1818	0.1818	Log-Log
CHOCOCHIP_FRAPPE	0.0202	0.0202	0.0202	Log-Log
VANILLA_LATTE	0.1414	0.1414	0.1414	Log-Log
CLASSIC_CARAMEL_MACCHIATO	0.303	0.303	0.303	Square Root

8.9 Seasonality

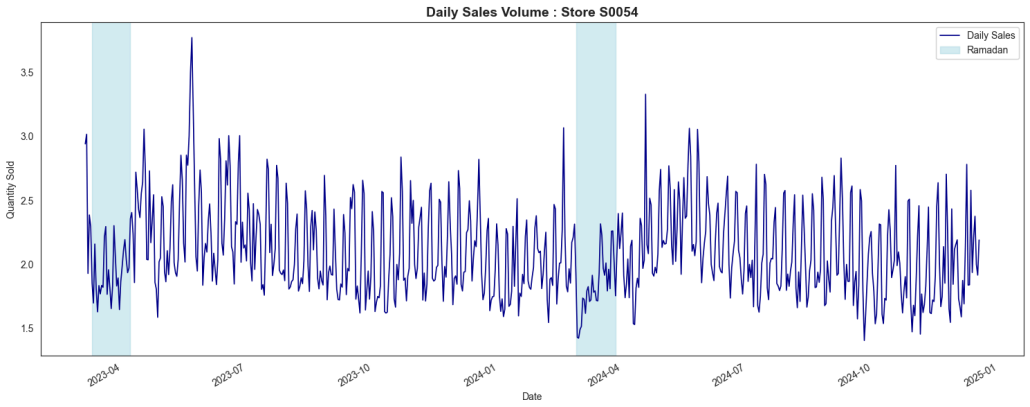


Figure 25: Daily Sales Trend with Ramadan Seasonality

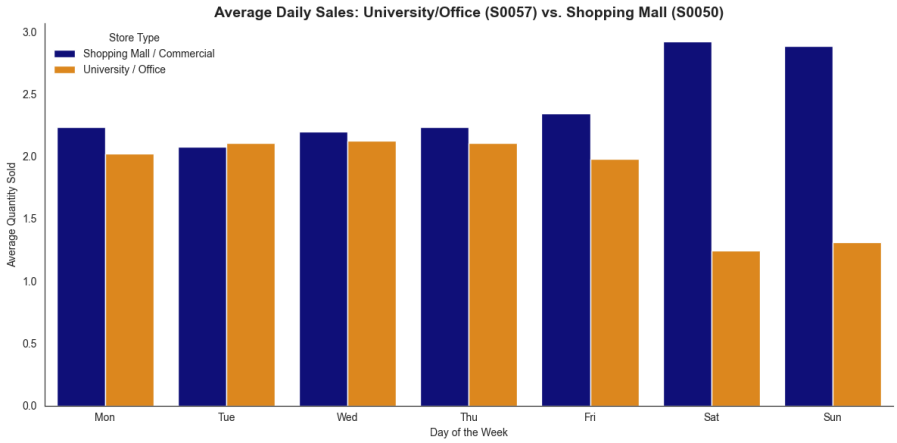


Figure 26: Weekly Seasonality by Venue

9 Appendix: Technical details of Log-Log linear demand model

9.1 Demand Curve

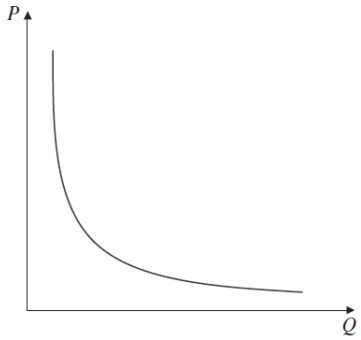


Figure 27: Log-log linear demand curve, from [9]

9.2 OLS Assumptions for the Demand Models

To ensure that the OLS estimators $\hat{a}_j, \hat{b}_{jk}$, and $\hat{\gamma}_j$ are the Best Linear Unbiased Estimators (BLUE), the following assumptions must hold for each equation j in the system:

1. **Linearity & Strict Exogeneity (OLS.1)**: The conditional mean of the log-quantity is a linear function of the log-prices and other covariates.

$$E(\ln Q_t \mid \ln P, \boldsymbol{X}, \alpha) = a + \sum_{k=1}^{J} b_k \ln P_{kt} + \boldsymbol{X}_t \delta + \alpha \tag{13}$$

   Otherwise written as the conditional mean of the error being zero given the entire history of the stochastic regressors:

$$E(\xi_t \mid \ln P, \boldsymbol{X}, \alpha) = 0 \tag{14}$$

2. **Stochastic Regressors (OLS.2)**: The explanatory variables $\{\ln P_{1t}, ..., \ln P_{Jt}, \boldsymbol{X}_t\}$ are random variables determined by market forces and environmental factors.

3. **Homoskedasticity (OLS.3)**: The conditional variance of the demand shock ξ is constant across all time periods.

$$\mathrm{Var}(\xi_t \mid \ln P, \boldsymbol{X}, \alpha) = \sigma^2 \tag{15}$$

4. **Conditional Independence (OLS.4)**: Given the regressors, the demand shocks ξ_t are independent across time, implying no serial correlation.

$$\xi_t \mid \ln P, \boldsymbol{X}, \alpha \quad \text{are independent for all } t \tag{16}$$

5. **Conditional Normality (OLS.5)**: The log-quantity follows a normal distribution conditional on the regressors.

$$\ln Q_t \mid \ln P, \boldsymbol{X}, \alpha \sim N\Big(a + \sum b_k \ln P_{kt} + \boldsymbol{X}_t\delta + \alpha, \sigma^2\Big) \tag{17}$$

6. **Identifiability / Full Rank (OLS.6)**: The matrix of stochastic prices and covariates must have full column rank with probability 1.

$$P(\mathrm{rank}(\boldsymbol{M}) = K) = 1 \tag{18}$$

where $\boldsymbol{M}$ is the matrix containing the intercept, all J prices, and covariates $\boldsymbol{X}$.

9.3 OLS Regression Estimates

Product	Rama	Pay_A	Pay_C	Pay_G	t	Lag1	Lag7	sin	cos	Adj R^2	W. R^2	AIC	BIC
CHEESE	−0.816***	–	−0.210	–	−0.0010***	0.234***	0.099***	0.183***	−0.059***	0.521	0.456	1776.0	1845.7
CAFFE	−0.436***	–	−0.258	–	−0.0005***	0.153***	0.171***	0.265***	−0.029*	0.729	0.293	1988.6	2054.2
VANILLA	−0.489***	–	–	–	−0.0009***	0.133***	–	0.185***	−0.100***	0.399	0.126	2641.4	2694.6
PECAN	−0.804***	–	−0.698**	–	−0.0011***	0.101***	0.053.	0.161***	−0.052**	0.425	0.122	2379.8	2437.2
HAZEL	−0.435***	–	–	–	−0.0007***	0.164***	0.123***	0.193***	−0.043*	0.360	0.219	2634.6	2692.8
COCOA	−0.627***	–	–	–	−0.0006***	0.118***	0.089**	0.194***	−0.076***	0.293	0.117	2684.1	2737.1
MATCHA	−0.445***	–	−0.555*	–	−0.0005***	0.162***	0.099***	0.154***	−0.042*	0.463	0.142	2439.6	2502.3
SALTED	−0.673***	0.317	–	–	−0.0009***	0.089***	0.051.	0.120***	−0.098***	0.282	0.172	2641.7	2705.3
CLASSIC	−0.738***	–	−0.559**	–	−0.0014***	0.111***	0.131***	0.151***	−0.104***	0.357	0.200	2444.8	2514.3
AMERI	−0.134	–	–	–	0.0004**	0.208***	0.150***	0.349***	−0.005	0.694	0.274	2771.0	2825.7

Table 22: OLS Control Variables and Model Fit by Product (Work)

Product	Rama	Pay_G	Pay_C	Pay_A	Lag1	Lag7	sin	cos	Adj R^2	W. R^2	AIC	BIC
CHEESE	−0.217***	–	–	−1.000***	0.381***	0.244***	−0.101***	−0.085***	0.669	0.455	16893.3	17185.1
CAFFE	−0.187***	–	–	−0.769***	0.192***	0.232***	−0.111***	−0.030***	0.663	0.275	14709.7	15007.5
VANILLA	−0.215***	−0.455***	−0.348***	−1.810***	0.148***	0.155***	−0.146***	−0.035***	0.352	0.206	29224.8	29533.5
PECAN	−0.283***	−0.639***	−0.473***	−1.190***	0.174***	0.151***	−0.149***	−0.054***	0.380	0.187	28718.5	29027.8
HAZEL	−0.175***	−0.380***	–	−1.600***	0.182***	0.178***	−0.147***	−0.036***	0.373	0.236	29496.8	29799.3
COCOA	−0.226***	−0.199**	−0.274***	−1.440***	0.124***	0.126***	−0.166***	−0.032***	0.299	0.137	30428.6	30738.6
MATCHA	−0.203***	−0.134.	−0.461***	−1.260***	0.145***	0.158***	−0.144***	−0.050***	0.371	0.167	30501.7	30804.3
SALTED	−0.196***	–	−0.522***	−1.510***	0.208***	0.213***	−0.161***	−0.051***	0.428	0.287	30601.8	30906.5
CLASSIC	−0.253***	–	−0.377***	−1.620***	0.226***	0.193***	−0.140***	−0.050***	0.475	0.286	26931.9	27244.1
AMERI	−0.090***	–	−0.176***	−0.250***	0.147***	0.164***	−0.109***	−0.022***	0.632	0.099	26173.5	26479.7

Table 23: OLS Control Variables and Model Fit by Product (Leisure)

9.4 2SLS Assumptions for the Single-Product Demand Equation

The structural model for a single product across stores s and time t is:

$$\ln Q_{st} = a + \sum_{k=1}^{J} b_k \ln P_{kst} + \boldsymbol{X}_{st}\delta + \alpha_s + \xi_{st} \tag{19}$$

1. **Linear in Parameters (2SLS.1)**: The model is linear in the parameters a, b_k, δ, and α_s. The instrumental variables are denoted as Z_{st}.
2. **Random Sampling (2SLS.2)**: We have a random sample of transaction data (quantity, prices, and controls) across the stores in Selangor.
3. **Rank Condition (2SLS.3)**:
   - (i) No perfect linear relationships exist among instruments Z_{st}.
   - (ii) The instruments Z_{st} must be sufficiently correlated with the endogenous prices $\ln P$ after controlling for $\boldsymbol{X}$ and α_s.

$$\text{Cov}(Z_{st}, \ln P_{kst} \mid \boldsymbol{X}_{st}, \alpha_s) \neq 0 \tag{20}$$

4. **Exogenous Instrumental Variables (2SLS.4)**: The error term ξ_{st} has a zero mean, and each instrument z is uncorrelated with the idiosyncratic demand shock.

$$E(\xi_{st}) = 0 \quad \text{and} \quad \text{Cov}(Z_{st}, \xi_{st}) = 0 \tag{21}$$

5. **Homoskedasticity (2SLS.5)**: The variance of the demand shock is constant, conditional on the instruments.

$$E(\xi_{st}^2 \mid Z) = \sigma^2 \tag{22}$$

6. **No Serial Correlation (2SLS.6)**: The demand shocks ξ_{st} are uncorrelated over time for a given store.

$$\text{Cov}\big(\xi_{st}, \xi_{s,t-1}\big) = 0 \tag{23}$$

9.5 Instrumental Variable Diagnostic Test

9.5.1 Weak Instrument Test

The weak instrument test evaluates whether the instruments are sufficiently correlated with the endogenous regressors. Weak instruments can lead to biased and inconsistent 2SLS estimates, potentially performing worse than ordinary least squares (OLS).

This test is based on the first stage regression:

$$\ln P_{kst} = Z_{st}\pi + \boldsymbol{X}_{st}\delta + \alpha_s + v_{st} \tag{24}$$

The strength of the instruments is assessed using the F-statistic:

$F = \frac{\frac{R^2}{q}}{\frac{1-R^2}{N-K}}$

where R^2 us the coefficient of determination from the first stage regression, q is the number of instruments, N is the sample size, and K is the total number of parameters.

- Null hypothesis (H_0): Instruments are weak.
- Alternative hypothesis (H_1): Instruments are strong.

A commonly used rule of thumb is that $F > 10$ indicates sufficiently strong instruments.

9.5.2 Wu-Hausman Test (Endogeneity Test)

The Wu-Hausman test evaluates whether the endogenous regressors are truly endogenous, and hence whether IV estimation is necessary.

The test compares OLS and 2SLS estimators:

$$H = \left(\hat{\beta}_{\text{IV}} - \hat{\beta}_{\text{OLS}}\right)^T \left[\text{Var}\left(\hat{\beta}_{\text{IV}}\right) - \text{Var}\left(\hat{\beta}_{\text{OLS}}\right)\right]^{-1} \left(\hat{\beta}_{\text{IV}} - \hat{\beta}_{\text{OLS}}\right) \tag{25}$$

Under the null hypothesis, the statistic follows an F-distribution or chi-squared distribution depending on implementation.

- Null hypothesis (H_0): Regressors are exogenous (OLS is consistent).
- Alternative hypothesis (H_1): Regressors are endogenous (IV/2SLS is required).

A statistically significant result indicates that OLS is inconsistent and supports the use of instrumental variables.

9.5.3 2SLS Estimator and R^2

9.5.3.1 Estimator

The 2SLS estimator for the parameter vector $\theta = (a, b_1, ..., b_J, \delta^T)^T$ is defined as:

$$\hat{\theta}_{\text{2SLS}} = (\boldsymbol{W}^T \boldsymbol{P}_{\boldsymbol{Z}} \boldsymbol{W})^{-1} \boldsymbol{W}^T \boldsymbol{P}_{\boldsymbol{Z}} \ln \boldsymbol{Q} \tag{26}$$

where:

- $\boldsymbol{W}$ is the matrix of regressors, including the endogenous log prices $\ln P_{kst}$ and exogenous controls $\boldsymbol{X}_{st}$.
- $\boldsymbol{P}_{\boldsymbol{Z}} = \boldsymbol{Z}(\boldsymbol{Z}^T \boldsymbol{Z})^{-1} \boldsymbol{Z}^T$ is the projection matrix (or "hat" matrix) that projects the variables in $\boldsymbol{W}$ onto the space spanned by the instruments $\boldsymbol{Z}$.
- $\boldsymbol{Z}$ contains the Hausman instruments $\overline{\ln P}_{k,-s,t}$ and the exogenous covariates $\boldsymbol{X}_{st}$.

9.5.3.2 Second Stage R^2

$$R^2 = 1 - \frac{RSS}{TSS} = 1 - \frac{\sum_{s,t} \left(\ln Q_{st} - \boldsymbol{W}_{st} \hat{\theta}_{\text{2SLS}}\right)^2}{\sum_{s,t} \left(\ln Q_{st} - \overline{\ln Q}\right)^2} \tag{27}$$

9.6 Model Diagnostics

10 Appendix: Technical Details of the CatBoost Model

This appendix provides additional technical details on the CatBoost algorithm and the post-estimation interpretation tools used in this study. In particular, it discusses CatBoost's treatment of categorical variables through ordered target statistics, the role of ordered boosting, and the interpretation of partial dependence plots (PDP), individual conditional expectation (ICE) plots, and SHAP values. The main methodology section focuses on the empirical implementation, while this appendix provides the algorithmic intuition supporting that implementation. CatBoost is an open-source gradient boosting library for decision trees with built-in support for numerical and categorical variables catboost.

10.1 CatBoost: Gradient Boosting with Symmetric Trees

For each target beverage, the CatBoost model uses store-day features that broadly mirror the linear demand model. Table 24 catalogues the complete feature set. Although CatBoost does not impose a parametric demand equation, it is estimated on a feature set closely aligned with the economic and operational controls used in the linear model — augmented

with purpose-level interactions and competitive-density measures — while allowing the fitted demand surface to remain flexible and nonlinear.

CatBoost constructs a predictive function as an additive ensemble of symmetric (oblivious) decision trees. The fitted model after M boosting iterations is

$$\hat{f}(x) = \sum_{m=1}^{M} \eta \cdot h_m(x) \tag{28}$$

where $h_m(x)$ is the m-th symmetric decision tree and η is the learning rate (shrinkage parameter) that controls the contribution of each tree to the ensemble. At each iteration m, the algorithm fits a new tree to the negative gradient of the loss function evaluated at the current ensemble's predictions. For the RMSE loss function used in this study, the negative gradient for observation i at iteration m reduces to the residual:

$$g_i^m = -\frac{\partial L\big(y_i, \hat{f}(x_i)\big)}{\partial \hat{f}(x_i)}) \mid_{\hat{f}=\hat{f}_{m-1}} = y_i - \hat{f}_{m-1}(x_i) \tag{29}$$

where $\hat{f}_{m-1}(x_i) = \sum_{k=1}^{m-1} \eta \cdot h_k(x_i)$ is the prediction of the ensemble after $m-1$ iterations. Each new tree h_m is fitted to approximate these residuals, and the ensemble is updated as

$$\hat{f}_m(x) = \hat{f}_{m-1}(x) + \eta \cdot h_m(x). \tag{30}$$

Symmetric (oblivious) trees. Unlike standard decision trees, CatBoost uses symmetric trees in which the same splitting feature and threshold are applied to all nodes at a given depth level. A symmetric tree of depth d has exactly 2^d leaf nodes, and every observation follows a path determined by the same sequence of d binary tests. This structure acts as an implicit regulariser: it restricts the hypothesis space relative to unconstrained trees of the same depth, reducing overfitting while maintaining computational efficiency (Prokhorenkova et al., 2018).

10.2 Catboost features table

Category	Feature(s)	Description
Own price	`log_up_target`	Log of the target product's effective unit price
	`discount_rate_target`	Discount rate of the target product (discount amount / gross sales)
Price × Purpose interactions	`log_up_target_x_{purpose}`	Own log price interacted with venue purpose (*Leisure*, *Work*, *Other*); captures purpose-heterogeneous price sensitivity
Demand lags	`log_qty_target_lag1`, `log_qty_target_lag2`, `log_qty_target_lag7`	Log of the target product's lagged daily sales quantity at horizons of 1, 2, and 7 days
Calendar & seasonal controls	`dow_sin`, `dow_cos`	Cyclical (sine / cosine) encoding of day-of-week
	`month_factor`	Month as a categorical variable (1–12)
	`t`	Linear time trend (days since first observation)
	`Ramadan2023`, `Ramadan2024`, `hol_Christmas_Day`	Binary indicators for Ramadan windows and public holidays
Store & venue identifiers	`store_code`	Store identity (categorical)
	`purpose`	Venue purpose category (*Leisure*, *Work*, *Other*); included as a categorical feature in the pooled model
	`Purpose_Stores_Active`	Count of daily active stores within the same purpose segment; proxies for same-segment competitive intensity
Payment composition	`pay_share_{method}`	Store-day share of transactions by payment method (e.g. e-wallet, credit card), excluding the base category (cash)
Rival product prices	`log_up_{rival}`	Log effective unit price of the top-k rival products (up to 8), selected by total sales volume
	`log_up_{rival}_x_{purpose}`	Rival log price interacted with venue purpose; allows cross-price effects to differ by store segment

Table 24: Complete feature set for the CatBoost demand model. Categorical features are passed directly to CatBoost's native categorical handler; all other features enter as numeric. The per-purpose model omits the `purpose` categorical and the price × purpose interaction terms, as each model is already estimated on a single-purpose subset.

10.3 Catboost Diagnostic Metrics

Product	Segment	R^2 (PP)	R^2 (Pool)	RMSE (PP)	RMSE (Pool)	MAPE (PP)	MAPE (Pool)	Δ R^2	Δ RMSE%
Cheese-cream Latte	Leisure	0.790	0.779	0.263	0.270	21.0	21.6	+0.012	+2.7%
Cheese-cream Latte	Work	0.921	0.668	0.145	0.297	11.5	24.3	+0.252	+51.2%
Caffè Latte	Leisure	0.699	0.780	0.311	0.266	25.5	21.5	–0.081	–16.8%
Caffè Latte	Work	0.932	0.892	0.187	0.235	14.7	18.7	+0.040	+20.7%
Americano	Leisure	0.728	0.721	0.397	0.402	33.7	34.2	+0.007	+1.3%
Americano	Work	0.966	0.832	0.161	0.358	12.8	29.9	+0.135	+55.1%
Matcha Latte	Leisure	0.760	0.692	0.348	0.394	28.9	33.6	+0.068	+11.8%
Matcha Latte	Work	0.838	0.713	0.311	0.414	26.1	36.1	+0.125	+24.9%
Classic Caramel Mac.	Leisure	0.604	0.610	0.411	0.407	35.7	35.3	–0.007	–0.9%
Classic Caramel Mac.	Work	0.632	0.508	0.371	0.428	31.7	38.2	+0.124	+13.5%
Salted Caramel Mac.	Leisure	0.580	0.576	0.451	0.453	39.8	40.1	+0.005	+0.6%
Salted Caramel Mac.	Work	0.751	0.458	0.332	0.490	27.1	43.6	+0.293	+32.2%
Hazelnut Latte	Leisure	0.717	0.699	0.367	0.378	30.6	31.5	+0.017	+2.9%
Hazelnut Latte	Work	0.788	0.688	0.308	0.374	25.2	31.2	+0.099	+17.4%
Pecan Crunch Latte	Leisure	0.623	0.630	0.422	0.419	36.7	36.3	–0.007	–0.9%
Pecan Crunch Latte	Work	0.900	0.641	0.230	0.435	18.6	38.0	+0.259	+47.2%
Vanilla Latte	Leisure	0.698	0.701	0.386	0.383	32.2	31.9	–0.004	–0.6%
Vanilla Latte	Work	0.826	0.703	0.287	0.375	23.6	31.3	+0.123	+23.4%
Cocoa Latte	Leisure	0.661	0.661	0.394	0.394	33.6	33.6	–0.001	–0.1%
Cocoa Latte	Work	0.650	0.602	0.385	0.411	33.1	35.0	+0.048	+6.2%

Table 25: Per-purpose (PP) vs combined model metrics, evaluated on segment-specific subsets. Both models estimated with Optuna-tuned hyperparameters. Log-scale RMSE and MAE; MAPE in percentage points.

10.4 Ordered Target Encoding for Categorical Features

CatBoost handles categorical features using *ordered target statistics*, a procedure designed to avoid target leakage during encoding (Prokhorenkova et al., 2018). For a categorical feature x_k^i observed for training example i, the encoded value is computed using only observations that precede i in a random permutation σ of the training set:

$$\hat{x}_k^i = \frac{\sum_{j:\sigma(j)<\sigma(i)} \mathbb{1}\left[x_k^j = x_k^i\right] \cdot y_j + a \cdot p}{\left(\sum_{j:\sigma(j)<\sigma(i)} \mathbb{1}\left[x_k^j = x_k^i\right] + a\right)} \tag{31}$$

where y_j is the target value (log quantity) for observation j, $\mathbb{1}[\cdot]$ is the indicator function, p is a prior (typically the global mean of the target), and $a > 0$ is a smoothing parameter that regularises the estimate toward the prior when few observations share the same categorical level.

The key property of this encoding is that the encoded value for observation i never uses its own target y_i or the targets of observations appearing later in the permutation. This eliminates the prediction shift that arises in standard target encoding — where the encoded value for a category is computed from all observations, including the one being encoded — and which is a form of target leakage that leads to overfitting, particularly for high-cardinality categorical features such as `store_code`.

In practice, CatBoost averages over multiple random permutations to reduce variance in the encoded values.

10.5 Ordered Boosting

Standard gradient boosting suffers from a subtle form of target leakage: the residuals g_i^m used to fit tree h_m are computed from $\hat{f}_{m-1}$, which was itself trained on all N observations including observation i. This means the residuals are biased — they are systematically smaller for training observations than for unseen data — leading to a *prediction shift* that compounds across boosting iterations (Prokhorenkova et al., 2018).

CatBoost addresses this through *ordered boosting*, which maintains a separate model for computing each observation's residual. For a random permutation σ of the training set, the residual for observation i at iteration m is computed from a model trained only on observations preceding i in the permutation:

$$g_i^m = y_i - \hat{f}_{m-1}^{\sigma(i)}(x_i) \tag{32}$$

where $\hat{f}_{m-1}^{\sigma(i)}$ denotes the ensemble after $m-1$ iterations trained exclusively on the set $\{j : \sigma(j) < \sigma(i)\}$ — that is, observations appearing before i in the permutation σ. This ensures that the residual for observation i is computed from a model that has never seen y_i, eliminating the prediction shift.

In full generality, maintaining N separate models (one per observation) would be computationally prohibitive. CatBoost approximates ordered boosting by maintaining $O(\log N)$ models corresponding to exponentially growing prefixes of the permutation and assigning each observation to the appropriate prefix model. The permutation is also re-sampled across iterations to reduce variance. The combination of ordered boosting with ordered target statistics (Appendix A.2) ensures that both the categorical encoding step and the gradient computation step are free of target leakage — a property that is unique to CatBoost among gradient boosting implementations and is particularly valuable when high-cardinality categorical features such as `store_code` are present.

10.6 Partial Dependence Plot (PDP)

The Partial Dependence Plot (Friedman, 2001) visualises the average marginal effect of a feature x_j on the model's prediction, marginalising over all other features x_C (the complement set). For a feature value x_j, the partial dependence function is defined as

$$\bar{f}(x_j) = \frac{1}{N} \sum_{i=1}^{N} \hat{f}(x_j, x_C^i) \tag{33}$$

where N is the number of observations in the dataset and x_C^i denotes the values of all features other than x_j for observation i. The PDP is computed by fixing x_j at each point on a grid, replacing x_j in all N observations with that grid value, generating predictions for all modified observations, and averaging the predictions.

In our implementation, the feature of interest is the log of own price ($\ln p$). The PDP price curve is computed over a grid of $G = 25$ raw price values, $p_1, ..., p_G$, spanning the 5th to 95th percentile of observed prices. For each grid value p_g:

$$\bar{Q}(p_g) = \frac{1}{N} \sum_{i=1}^{N} \exp\big(\hat{f}(\ln p_g, x_C^i)\big) \tag{34}$$

where the exponentiation converts the log-quantity prediction back to the quantity scale. The resulting curve $\bar{Q}(p_g)$ traces the average predicted demand as a function of price, holding all other features at their observed values.

10.7 Individual Conditional Expectation (ICE)

ICE plots (Goldstein et al., 2015) decompose the PDP into its individual observation-level components. While the PDP shows the *average* effect of a feature, ICE plots show the effect for each observation separately, revealing heterogeneity that the PDP average may mask.

For a single observation i and feature value x_j, the ICE function is

$$\hat{f}^i(x_j) = \hat{f}(x_j, x_C^i) \tag{35}$$

which is simply the model's prediction for observation i with feature x_j set to the specified value and all other features held at their observed values. The PDP is the average of the ICE curves:

$$\bar{f}(x_j) = \frac{1}{N} \sum_{i=1}^{N} \hat{f}^i(x_j). \tag{36}$$

In our implementation, a random sample of $n = 50$ store-day observations is drawn, and each observation's ICE curve is plotted over the same price grid used for the PDP. Crossing or diverging ICE lines indicate that the price effect is heterogeneous across stores — for example, some stores may exhibit price sensitivity at a level where others do not — while parallel ICE lines indicate a homogeneous price response that the PDP faithfully represents.

10.8 A.5 Revenue Optimisation via Grid Search

The revenue optimisation identifies the price p^* that maximises total predicted revenue for a given product within a given purpose segment. The procedure uses the fitted CatBoost demand model as a counterfactual simulator.

Price grid construction. Let $p_{0.05}$ and $p_{0.95}$ denote the 5th and 95th percentiles of the observed effective price distribution for the target product within the relevant segment. A grid of $G = 40$ candidate prices is constructed as

$$p_g = p_{0.05} + \frac{g-1}{G-1}(p_{0.95} - p_{0.05}), \quad g = 1, ..., G. \tag{37}$$

This restriction ensures that the model is not asked to predict demand at prices outside its training support. Tree-based models are particularly unsuited to extrapolation because predictions outside the observed range default to the nearest leaf value, yielding unreliable counterfactuals.

Counterfactual demand prediction. For each candidate price p_g, the log price feature is set to $\ln(p_g)$ for all N store-day observations in the segment, while all other features are held at their observed values:

$$\hat{Q}_{ist}(p_g) = \exp\big(\hat{f}(\ln p_g, x_{C,st})\big) \quad \forall (s,t) \in \mathcal{S} \tag{38}$$

where $\mathcal{S}$ is the set of all store-day observations in the segment and $x_{C,st}$ denotes the observed values of all non-price features for store s on day t.

Revenue evaluation. Total predicted revenue at candidate price p_g is

$$R(p_g) = p_g \cdot \sum_{(s,t)\in\mathcal{S}} \hat{Q}_{ist}(p_g). \tag{39}$$

Optimal price selection. The revenue-maximising price is

$$p^* = \arg \max_{p_g \in \{p_1, ..., p_G\}} R(p_g). \tag{40}$$

This grid-search approach is computationally straightforward and does not require the revenue function to be differentiable or concave — properties that are not guaranteed for tree-based models whose fitted surfaces are piecewise constant. The approach follows the predict-then-optimise paradigm established in the retail analytics literature (Ferreira, Lee, and Simchi-Levi, 2016), adapted to a CatBoost demand model with segment-specific estimation.

Extension to profit maximisation. If unit cost c_i is available, the objective can be replaced with

$$\pi(p_g) = (p_g - c_i) \cdot \sum_{(s,t)\in\mathcal{S}} \hat{Q}_{ist}(p_g) \tag{41}$$

and the optimal price becomes $p^* = \arg\max_{p_g} \pi(p_g)$. Since marginal cost data were not available in this study, the analysis maximises revenue as the objective.

10.8.1 Summary of the Technical Interpretation Strategy

Taken together, ordered target statistics and ordered boosting explain why CatBoost is suitable for store-level beverage demand modelling with many categorical variables, while PDP, ICE, and SHAP explain how the fitted model can be interpreted despite not producing simple regression coefficients. In the main report, CatBoost is therefore treated primarily as a flexible predictive model whose price-response behaviour is summarized through model-

implied demand curves and explanation plots, rather than through directly reported scalar elasticities.

10.9 Catboost Diagonstic plots

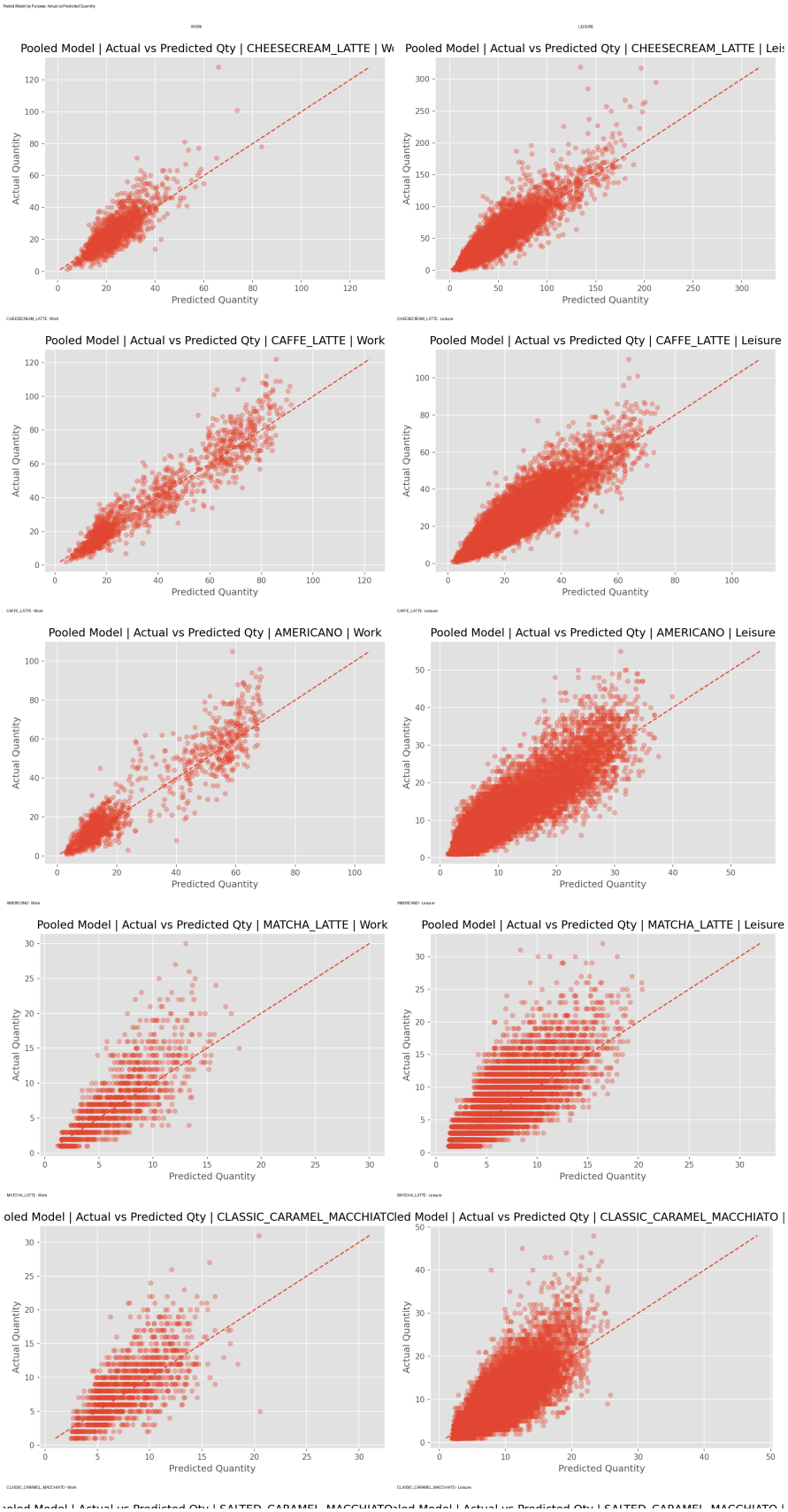


Figure 28: All Pred 1 Image

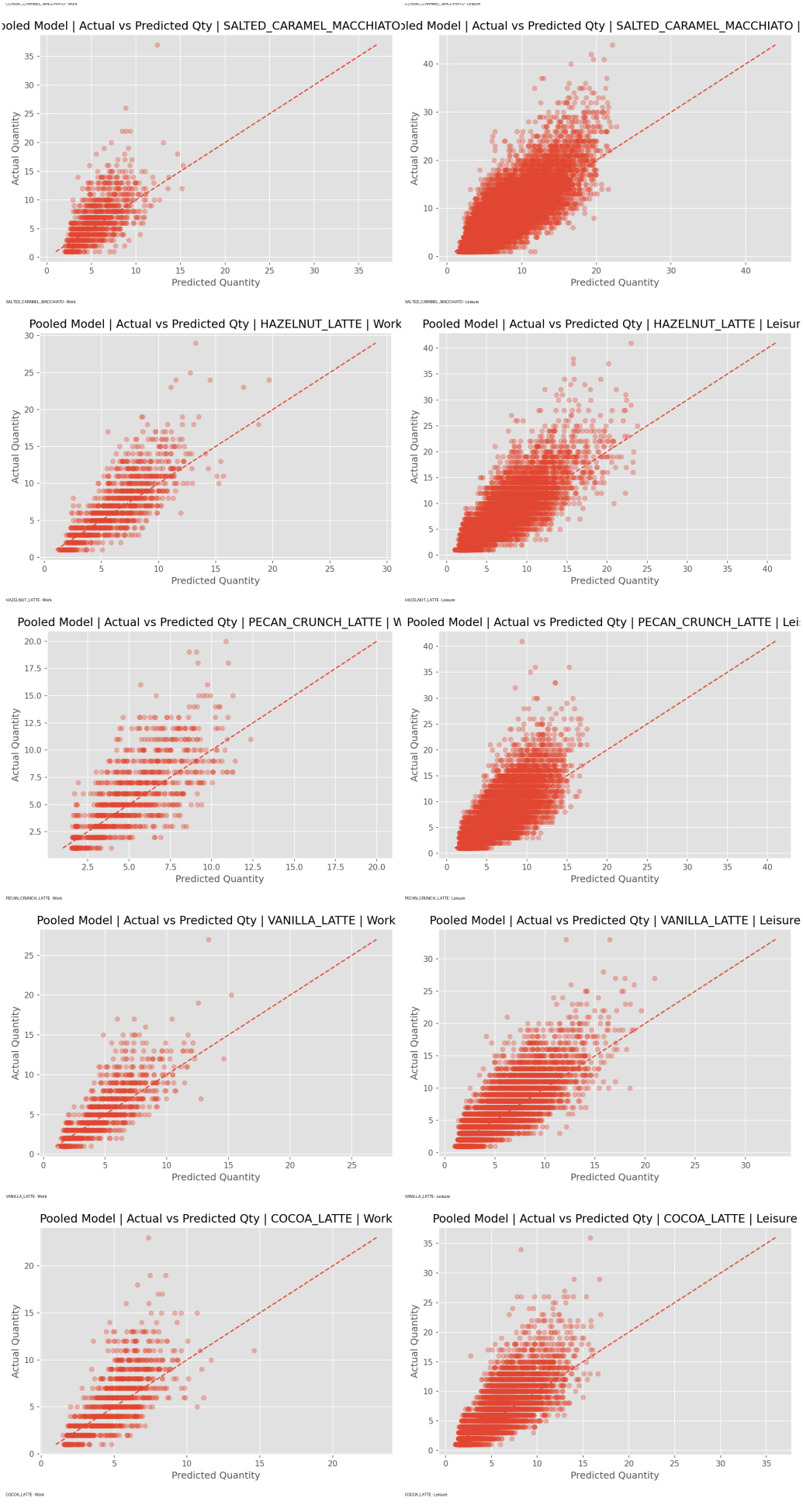


Figure 29: All Pred 2 Image

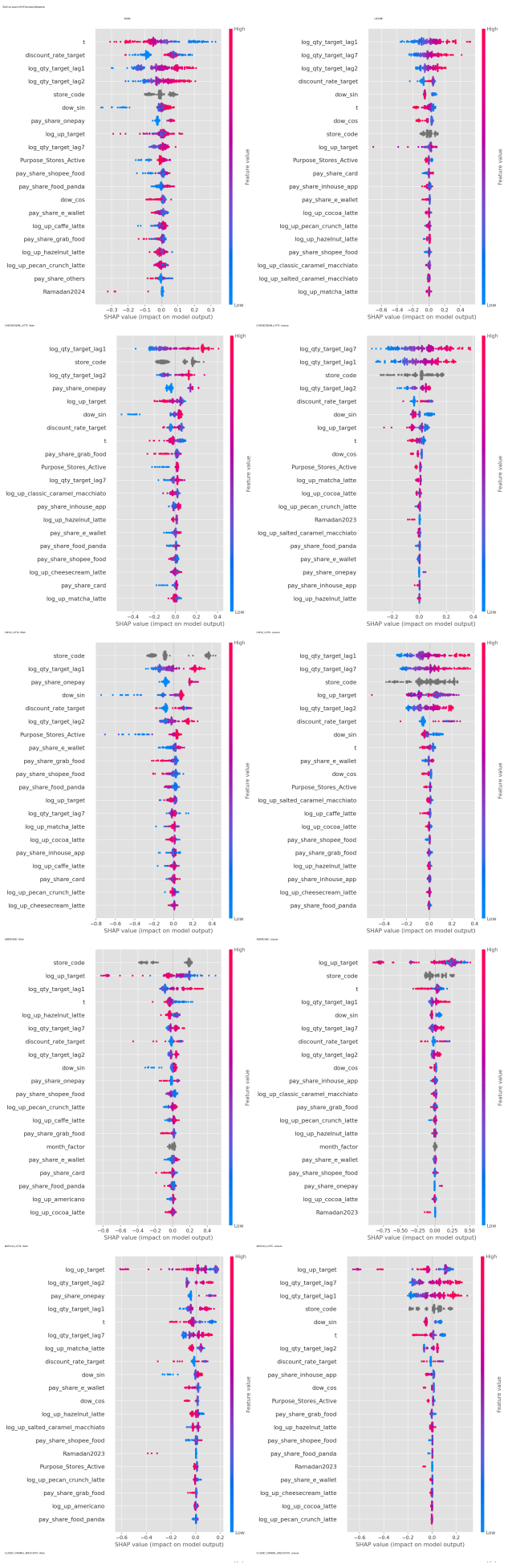


Figure 30: All SHAP 1 Image

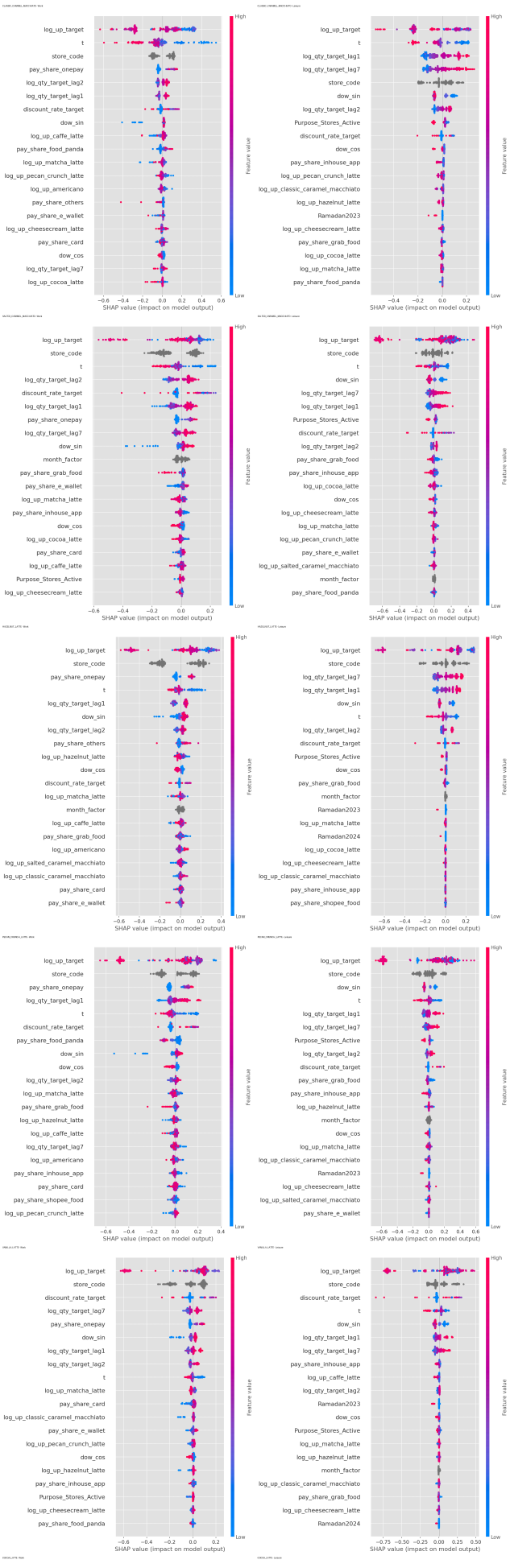


Figure 31: All SHAP 2 Image



Figure 32: All PDP 1 Image



Figure 33: All PDP 2 Image

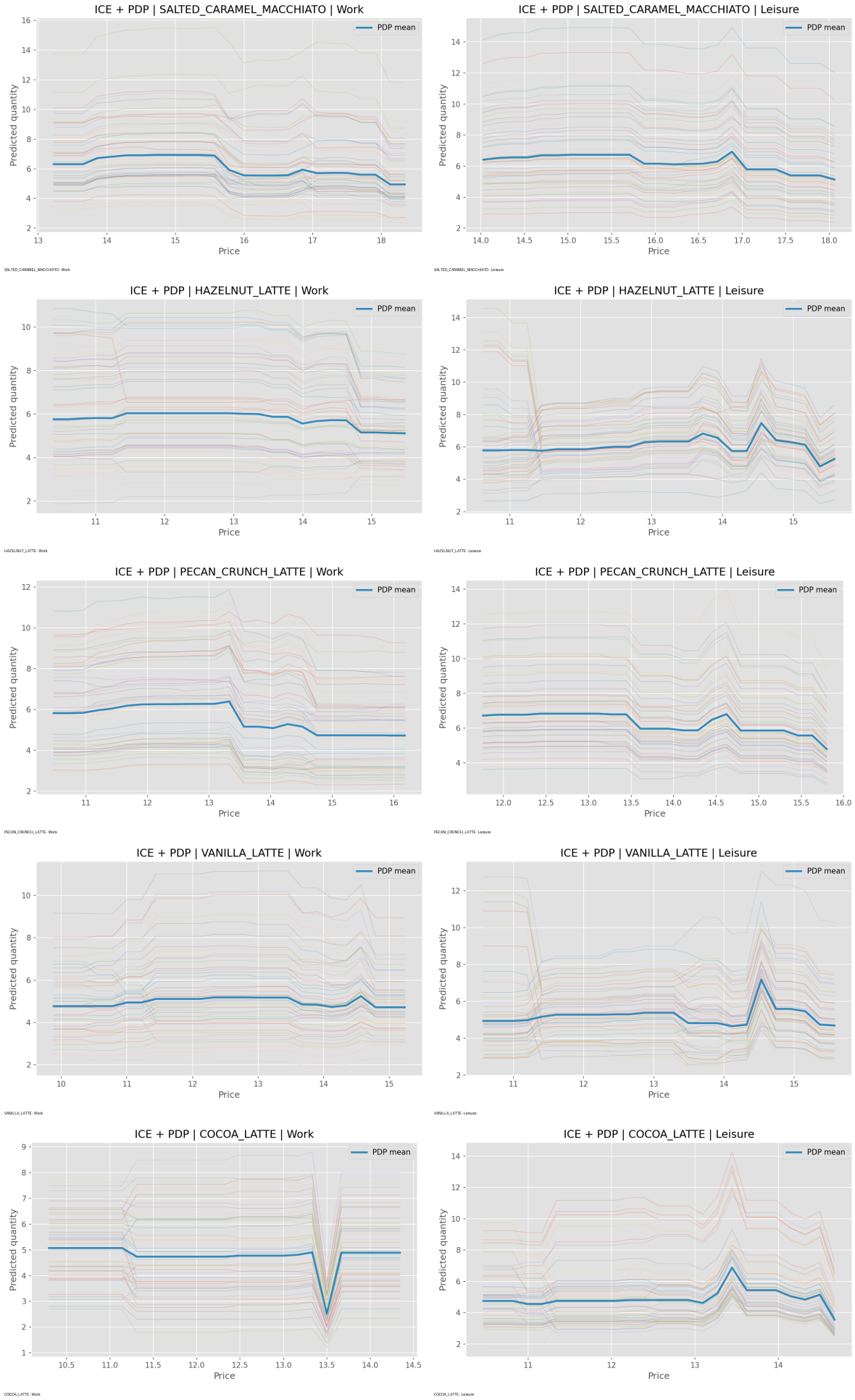


Figure 34: All ICE 2 Image

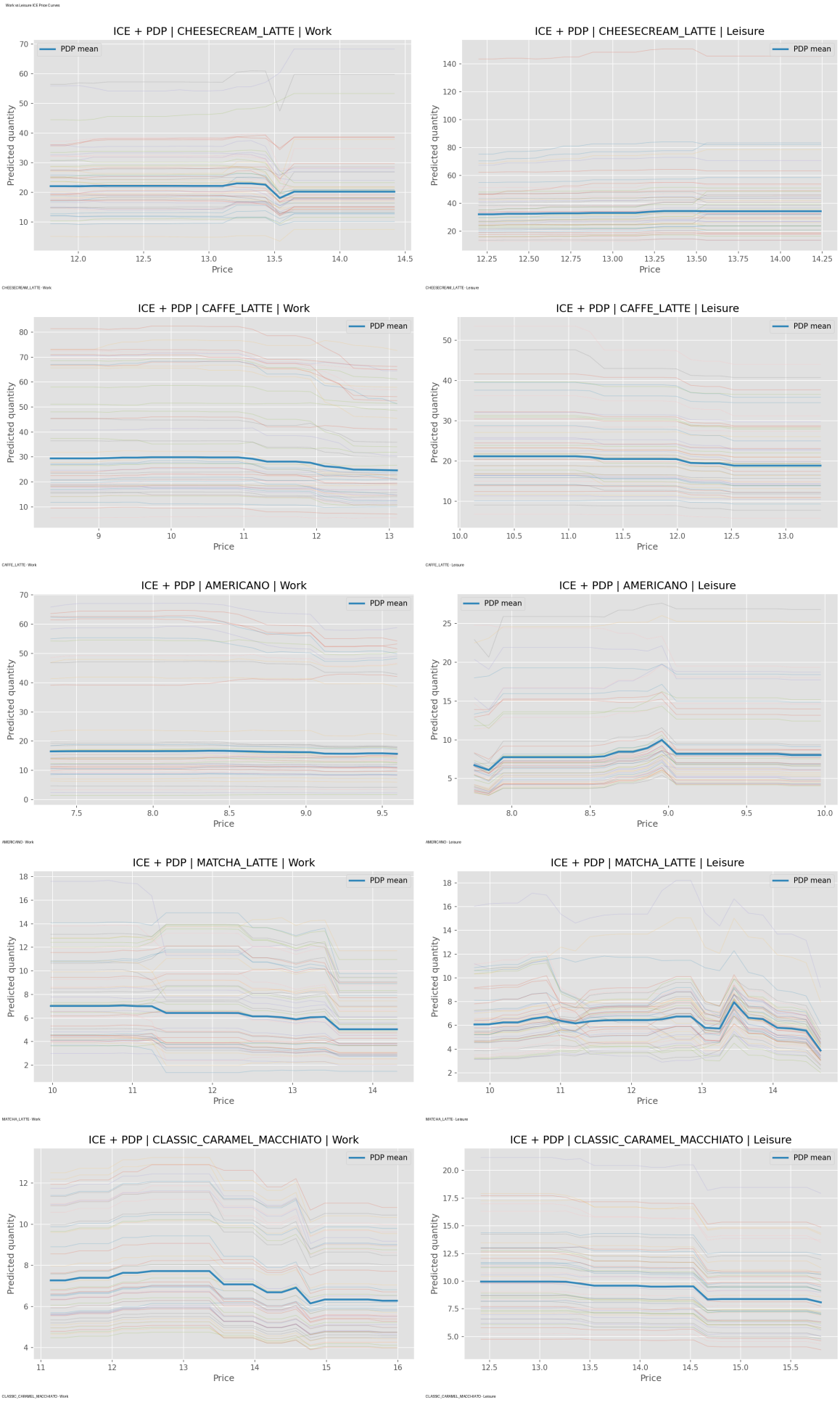


Figure 35: All ICE 1 Image

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